Constrained Gravity and Strong Fields

Constraint algebra, regular reduction, and spherical geometry

Geometry, constraints, and strong fields

Gauge covariance identifies compatible transport. Gravity asks for more: the geometry, constraints, and evolution must be specified together. A propagation rule, a canonical constraint algebra, and an evolved matter system are different mathematical objects, with different hypotheses.

This account proceeds from weighted transport geometry and reduced Gowdy evolution to finite linearized ADM constraints. The nonlinear obstruction explains why a nonzero derivative cannot satisfy the ordinary Leibniz rule on a complete finite pointwise algebra. Matrix moment maps, homological transfer, and spectral representations provide alternatives in their stated settings.

A real Lorentzian action supplies a classical gauge-theoretic parent. Boundary constructions and local second-class reduction retain their regularity and nondegeneracy conditions. The spherical sector makes mass, flux, and horizon diagnostics explicit, allowing exact identities to be compared with bounded numerical experiments.

The results do not amount to a complete continuum or quantum theory of gravity. A manufactured electrovac trajectory is not an existence theorem for nonlinear collapse. A scalar trapped-sphere diagnostic is not a continuum formation theorem without stability, convergence, and event control. These distinctions are part of the mathematics rather than qualifications added after it.

The shared resolution language continues in the arithmetic and polyphase constructions that follow. It makes the state spaces, comparison maps, and residual obstructions explicit without equating their physical interpretations.

Reading routes and conventions

For the geometric route, begin with transport and local frames, then follow the Gowdy model and the finite linearized ADM system. The nonlinear obstruction explains why a product rule that is harmless in a continuum calculation becomes a substantive issue on a finite pointwise algebra.

For the constrained-system route, begin with the canonical bracket and the exact finite moment map. Continue through homological transfer, resolved spectral evaluation, the classical BV parent, and the boundary and local reduction conditions. A field-dependent change of ghost variables and a canonical transformation of the extended phase space must be distinguished. The latter carries the appropriate momentum terms with it.

For the strong field route, begin with the spherical mass identity. The charged horizon calculations and the scalar mass-flux compatibility give concrete statements that can be checked independently. The numerical records then test selected trajectories and initial data. A manufactured gauge trajectory of a fixed spacetime and an evolved scalar-collapse model answer different questions.

Convention	Meaning
Canonical brackets	Each sector declares its variables and sign. In standard coordinates, $\{F, G\} = \sum_a (F_{q_a} G_{p_a} - F_{p_a} G_{q_a})$, so $\dot{F} = \{F, H\}$.
Geometry and signature	Lorentzian signature, curvature orientation, density weights, lapse, shift, and tetrad conventions are stated where they enter. Local transport alone does not select an Einstein action.
Gauge and reduction	Gauge choice, constraint enforcement, and removal of second-class variables are separate operations. An invertible gauge-fixing or Dirac matrix defines the regular chart used in the corresponding argument.
Spectral resolution	K denotes retained degree and N the sampling resolution in the resolved constructions. Conditions such as $16K < N$ are sufficient degree budgets in a declared representation. They are not invariant under every nonlinear change of variables.
Locality	A finite-range stencil, a full spectral evaluation, a matrix commutator, and a radial prefix constraint solve have different dependence on the data. The implementation is named before its locality is assessed.
Time stepping	The conservation proof for a projected discrete-gradient step concerns an autonomous Hamiltonian and an exactly solved implicit step. Approximate solves retain a residual that must be measured.
Spherical units	Geometrized units are used in the spherical examples. The mass, charge, radial coordinate, and slicing branch are defined before horizon or flux formulas are used.
Evidence	Ordinary proofs, inherited executable records, newly run bounded computations, and archived formal declarations retain their own scope. Numerical plots do not substitute for an existence or convergence proof.

Geometry and constrained evolution

1 Geometric transport and local work

The first gravitational question is local: how does a prescribed geometry change transport and energy exchange? A second question is dynamical: which equations determine that geometry? Keeping these questions distinct lets a finite model establish an exact balance law before it is coupled to a gravitational constraint system. Throughout this section the metric coefficients are supplied functions. The scalar field evolves on them.

1.1 A finite scalar action on a diagonal metric

Take a periodic chain with $N \geq 3$ sites, spacing $h > 0$, and indices modulo N. The continuum metric motivating its coefficients is

$$ds^2 = -N(t,x)^2 dt^2 + a(t,x)^2 dx^2, \qquad N(t,x) > 0, \quad a(t,x) > 0.$$

The repeated letter N in this metric is the lapse, whereas the integer N counts lattice sites. In the finite formulas below only the coefficient names w, c, and v are needed:

$$w_j = \frac{a_j}{N_j} > 0, \qquad c_{j+1/2} = \frac{N_{j+1/2}}{a_{j+1/2}} > 0, \qquad v_j = N_j a_j m^2 \geq 0.$$

Thus N/a is the coordinate characteristic speed of the continuum massless equation. Assume these finite coefficient arrays are continuously differentiable in time. For a real field $\varphi_j(t)$ set $\Delta_j = \varphi_{j+1} - \varphi_j$ and define

$$L_h = \frac{h}{2} \sum_j w_j \dot{\varphi}_j^2 - \frac{1}{2h} \sum_j c_{j+1/2} \Delta_j^2 - \frac{h}{2} \sum_j v_j \varphi_j^2. \tag{1}$$

Variations with compact time support give

$$w_j \ddot{\varphi}_j + \dot{w}_j \dot{\varphi}_j = \frac{c_{j+1/2}\Delta_j - c_{j-1/2}\Delta_{j-1}}{h^2} - v_j \varphi_j. \tag{2}$$

The neighboring edge terms in the action produce the two terms in the spatial numerator. The time integration by parts differentiates $w_j \dot{\varphi}_j$, including the $\dot{w}_j \dot{\varphi}_j$ term.

Proposition 1.1 (Local energy, flux, and geometric work)**.** *For every twice differentiable solution of* (2)*, define*

$$\begin{aligned} e_j &= \frac{h}{2} w_j \dot{\varphi}_j^2 + \frac{h}{2} v_j \varphi_j^2 + \frac{c_{j+1/2}}{4h} \Delta_j^2 + \frac{c_{j-1/2}}{4h} \Delta_{j-1}^2, \\ F_{j+1/2} &= -\frac{c_{j+1/2}}{2h} \Delta_j (\dot{\varphi}_j + \dot{\varphi}_{j+1}), \\ s_j &= -\frac{h}{2} \dot{w}_j \dot{\varphi}_j^2 + \frac{h}{2} \dot{v}_j \varphi_j^2 + \frac{\dot{c}_{j+1/2}}{4h} \Delta_j^2 + \frac{\dot{c}_{j-1/2}}{4h} \Delta_{j-1}^2. \end{aligned} \tag{3}$$

Then

$$\dot{e}_j + F_{j+1/2} - F_{j-1/2} = s_j. \tag{4}$$

In particular, $E_h = \sum_j e_j$ *satisfies* $\dot{E}_h = \sum_j s_j = -\partial_t L_h$*, where the partial derivative holds* φ *and* $\dot{\varphi}$ *fixed.*

Proof. Put $B_j = c_{j+1/2} \Delta_j$. Using the equation of motion, the derivative of the kinetic and mass terms at site j is

$$\frac{\dot{\varphi}_j}{h} (B_j - B_{j-1}) - \frac{h}{2} \dot{w}_j \dot{\varphi}_j^2 + \frac{h}{2} \dot{v}_j \varphi_j^2.$$

The right edge contribution differentiates to

$$\frac{\dot{c}_{j+1/2} \Delta_j^2}{4h} + \frac{B_j}{2h} (\dot{\varphi}_{j+1} - \dot{\varphi}_j),$$

and the left edge has the analogous expression. Adding these terms leaves $B_j(\dot{\varphi}_j + \dot{\varphi}_{j+1})/(2h)$ on the right and its negative counterpart on the left, plus s_j. These are exactly the flux terms in (4). Periodic summation cancels every interface flux. Each edge coefficient derivative appears twice with weight $1/(4h)$, giving $-\partial_t L_h$. □

Every e_j is nonnegative under the stated coefficient assumptions. An interval of cells $r, \dots, s$ obeys the exact coarse balance

$$\frac{d}{dt} \sum_{j=r}^{s} e_j + F_{s+1/2} - F_{r-1/2} = \sum_{j=r}^{s} s_j.$$

Thus coarse energy is obtained by adding fine cell energies and retaining only the exposed boundary fluxes. This is the same cancellation mechanism as projector transfer, applied here to a classical field energy. Its physical units and constitutive coefficients come from the action.

Example 1.2 (Time dependent expansion)**.** Let the lapse equal one, let $a(t) > 0$ be spatially uniform, and let $m = 0$. Then $w = a$, $c = 1/a$, and

$$\dot{E}_h = -\frac{h}{2} \dot{a} \sum_j \dot{\varphi}_j^2 - \frac{\dot{a}}{2ha^2} \sum_j \Delta_j^2.$$

An increasing a transfers energy out of this coordinate-time field Hamiltonian. The supplied background receives no evolution equation from this calculation. A self-consistent matter–geometry model must also evolve its gravitational variables.

For a smooth field sampled at sites and a smooth coefficient sampled at edge midpoints, the spatial expression is second-order consistent with $\partial_x (c \partial_x \varphi)$. Indeed, expanding at x_j gives

$$c(x_j + h/2)[\varphi(x_j + h) - \varphi(x_j)] - c(x_j - h/2)[\varphi(x_j) - \varphi(x_j - h)] = h^2 (c\varphi_x)_x(x_j) + O(h^4).$$

Uniform bounds on the requisite fourth derivatives make the remainder uniform on a compact spatial region. The exact finite balance holds without any Taylor approximation.

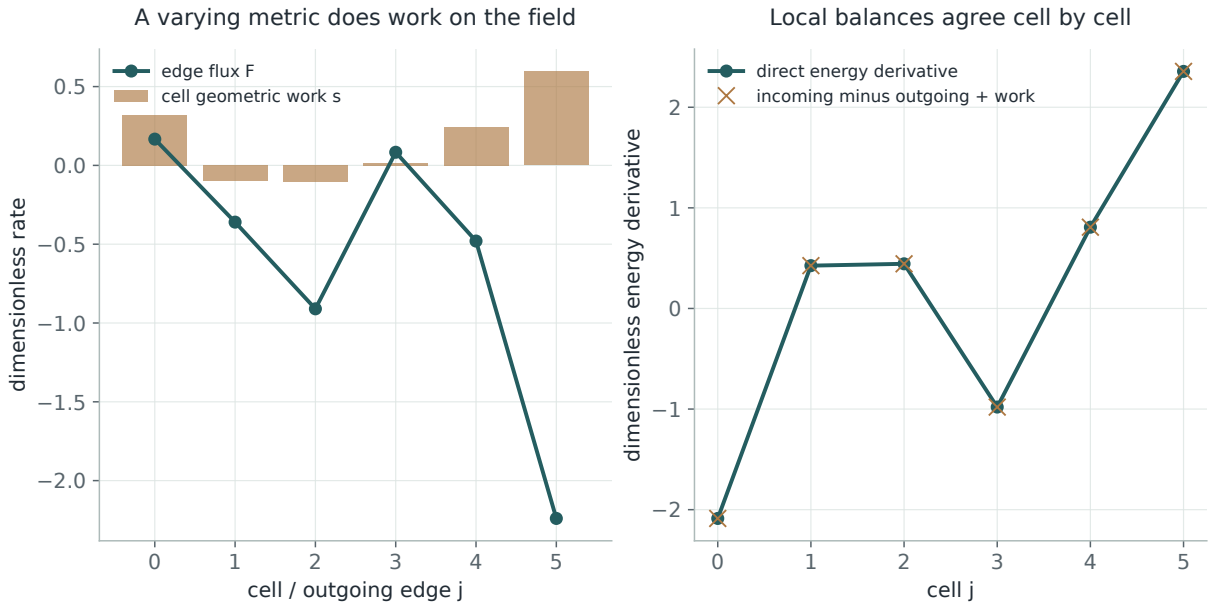


Figure 1. A six-cell example of prescribed metric work. Each direct cell-energy derivative agrees with incoming flux minus outgoing flux plus the coefficient work term. The curves illustrate the exact finite identity for a prescribed background.

1.2 Conformal curvature as an exact boost holonomy

Consider a twice continuously differentiable real function ϕ on an oriented (t, x) rectangle, and the coframe

$$e^0 = e^{\phi} dt, \qquad e^1 = e^{\phi} dx, \qquad ds^2 = e^{2\phi}(-dt^2 + dx^2).$$

Write

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \omega = (\phi_x dt + \phi_t dx) K.$$

The coframe satisfies the torsion-free Cartan equations with this connection. Since every connection value is a multiple of the same matrix, $\omega \wedge \omega = 0$. Therefore

$$\Omega = d\omega = (\phi_{tt} - \phi_{xx}) K \, dt \wedge dx, \qquad R = 2e^{-2\phi}(\phi_{tt} - \phi_{xx}). \tag{5}$$

This fixes the scalar-curvature sign convention used in this monograph.

We use the archive's plus-sign edge convention: for each oriented boundary edge define $U_e = \exp(+\int_e \omega)$. This specifies the signed transport observable used below. Multiplying in boundary order yields an exact identity

$$W_{\square} = \exp\left(K \oint_{\partial \square} (\phi_x dt + \phi_t dx) \right) = \exp\left(\frac{K}{2} \int_{\square} e^{2\phi} R \, dt \, dx \right). \tag{6}$$

To see the second equality directly, integrate ϕ_{tt} along the two horizontal endpoints and ϕ_{xx} along the two vertical endpoints, with the boundary orientation from $dt\wedge dx$. This is the rectangular fundamental theorem of calculus. No path-order correction survives because the edge exponents commute.

If the integrated coefficient is χ, then

$$e^{\chi K} = \cosh\chi\, I + \sinh\chi\, K.$$

This preserves the Lorentz form $\mathrm{diag}(-1,1)$ and has positive eigenvalues e^{χ} and $e^{-\chi}$. Its real principal matrix logarithm is χK. These are Lorentz boosts. The positive Hilbert norm used for unitary color transport in the gauge monograph is a different structure.

Example 1.3 (Constant curvature density)**.** For $\phi(t,x) = b(t^2-x^2)/2$, one has

$$\phi_{tt} - \phi_{xx} = 2b, \qquad R = 4b\, e^{-2\phi}, \qquad W_{\square} = e^{2b\ \mathrm{area}(\square)K}.$$

The curvature density $e^{2\phi}R/2$ is constant although R varies with position. Adding oriented subrectangle integrals reconstructs the coarse rapidity exactly.

The plus-sign observable is assigned the endpoint action

$$G(t,x) = e^{\chi(t,x)K}, \qquad U'_{a\to b} = G(b)U_{a\to b}G(a)^{-1}.$$

It corresponds to the signed connection rule $\omega'_{+} = \omega_{+} + d\chi\, K$. Intermediate endpoint factors cancel, so the loop observable is conjugated at its base point. For the Cartan convention $D = d + \omega$, a coframe transformation $e \mapsto Ge$ instead gives $\omega' = \omega - d\chi\, K$ and parallel transport $\exp(-\int_e \omega)$. Its closed-loop rapidity is the negative of the rapidity of $W_{\square}$ defined here. The fixed-frame curvature identity (6) is unchanged. Declaring these signs avoids mixing the archive's edge observable with a different Cartan parallel-transport convention.

2 Weak gravity as a sign and consistency benchmark

Weak gravity provides checks on attraction, variational signs, and optical observables. These checks are performed within the weak-field model. They do not furnish nonlinear gravitational evolution.

2.1 Radial Poisson equation and a finite variational problem

Let $G > 0$, let $\rho(r)$ be a continuous spherical density, and impose regularity $\lim_{r\downarrow 0} r^2\Phi'(r) = 0$. The radial Poisson equation gives

$$(r^2\Phi')' = 4\pi G r^2\rho, \qquad M_{\mathrm{enc}}(r) = 4\pi\int_0^r s^2\rho(s)\,ds, \qquad \Phi'(r) = \frac{GM_{\mathrm{enc}}(r)}{r^2}.$$

The acceleration is $a_r = -\Phi'$. A positive density therefore gives an inward radial acceleration.

Putting $u(r) = r\Phi(r)$ removes the radial first derivative:

$$r^2\Phi' = ru' - u, \qquad (r^2\Phi')' = ru'', \qquad u'' = 4\pi G r\rho.$$

For fixed endpoint values, this equation is the stationary condition of

$$S[u] = \int_0^{R_0} \left[\frac{(u')^2}{2G} + 4\pi r \rho u \right] dr.$$

On $r_i = ih$, $i = 0, \dots, M$, with $Mh = R_0$, its finite counterpart is

$$S_h[u] = \sum_{i=0}^{M-1} \frac{(u_{i+1} - u_i)^2}{2Gh} + 4\pi h \sum_{i=1}^{M-1} r_i \rho_i u_i. \tag{7}$$

Proposition 2.1 (Unique finite radial solution)**.** *With u_0, u_M fixed, S_h has a unique minimizer. Its interior values satisfy*

$$\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} = 4\pi G r_i \rho_i.$$

Proof. Differentiation with respect to u_i gives $(2u_i - u_{i-1} - u_{i+1})/(Gh) + 4\pi h r_i \rho_i$. Setting it to zero gives the equation. For a variation v with $v_0 = v_M = 0$, the Hessian quadratic form is

$$\frac{1}{Gh} \sum_{i=0}^{M-1} (v_{i+1} - v_i)^2.$$

Its vanishing forces v to be constant and hence zero. The finite Hessian is positive definite, so the quadratic action has one stationary point and that point is its unique minimizer. □

The boundary value $u_0 = 0$ allows a regular finite potential at the origin. The quotient u/r is used only for $r > 0$, with its continuous limit determined separately at $r = 0$.

For the normalized Gaussian density

$$\rho(r) = \frac{M_0}{(2\pi)^{3/2}\sigma^3} e^{-r^2/(2\sigma^2)}, \qquad \sigma > 0,$$

direct differentiation verifies

$$M_{\text{enc}}(r) = M_0 \left[\operatorname{erf}\left(\frac{r}{\sqrt{2}\sigma} \right) - \sqrt{\frac{2}{\pi}} \frac{r}{\sigma} e^{-r^2/(2\sigma^2)} \right],$$
$$\Phi(r) = -\frac{GM_0}{r} \operatorname{erf}\left(\frac{r}{\sqrt{2}\sigma} \right).$$

The potential tends to zero at infinity and has the finite central value $-\sqrt{2/\pi}\, GM_0/\sigma$. This removes the point-source singularity while preserving the attractive sign.

2.2 Clocks and straight-ray optical integrals

In the stipulated first-order isotropic weak-field metric,

$$ds^2 = -\left(1 + \frac{2\Phi}{c_0^2}\right) c_0^2 dt^2 + \left(1 - \frac{2\Phi}{c_0^2}\right) d\mathbf{x}^2, \qquad \left|\frac{\Phi}{c_0^2}\right| \ll 1,$$

a static clock has lapse $\mathcal{N} = \sqrt{1 + 2\Phi/c_0^2}$. The static-frequency ratio in this approximation is

$$\frac{\nu_{\text{obs}}}{\nu_{\text{emit}}} = \frac{\mathcal{N}_{\text{emit}}}{\mathcal{N}_{\text{obs}}} = 1 + \frac{\Phi_{\text{emit}} - \Phi_{\text{obs}}}{c_0^2} + O\left(\frac{\Phi^2}{c_0^4}\right).$$

For $\Phi(x, b) = -GM_0/\sqrt{x^2 + b^2}$, $b > 0$, evaluating the first-order straight-ray integrals over $[-L, L]$ gives

$$\Delta t_{\text{Shapiro}} = -\frac{2}{c_0^3}\int_{-L}^{L} \Phi(x, b)\, dx = \frac{4GM_0}{c_0^3}\,\text{asinh}\left(\frac{L}{b}\right),$$
$$\alpha(L, b) = \frac{2}{c_0^2}\left|\int_{-L}^{L} \partial_b \Phi(x, b)\, dx\right| = \frac{4GM_0 L}{c_0^2 b\sqrt{L^2 + b^2}}.$$

Here α is the magnitude of the deflection toward the source. The limit $L \to \infty$ gives $4GM_0/(c_0^2 b)$. The displayed equalities evaluate the weak-field integrals exactly. The approximation to the physical ray already entered through the metric expansion and the undeviated integration path.

3 Gowdy evolution and reconstruction of the remaining constraint

The next step is an inhomogeneous nonlinear model with two gravitational field channels. We take the archived Gowdy reduction as specified by its action and constraint densities. The results below prove the internal variational and reconstruction identities of that reduced system.

Let P, Q be real smooth fields on $I \times \mathbb{T}$, with $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ and $\gamma(\tau) = e^{-2\tau}$. Define

$$S_G = \frac{1}{2}\int_I \int_0^{2\pi} \left[P_\tau^2 - \gamma P_\theta^2 + e^{2P}(Q_\tau^2 - \gamma Q_\theta^2)\right] d\theta\, d\tau. \tag{8}$$

Time variations vanish at the endpoints of I and spatial variations are periodic. Integration by parts gives the two Euler–Lagrange expressions

$$\mathcal{E}_P = P_{\tau\tau} - \gamma P_{\theta\theta} - e^{2P}(Q_\tau^2 - \gamma Q_\theta^2),$$
$$\mathcal{E}_Q = Q_{\tau\tau} - \gamma Q_{\theta\theta} + 2(P_\tau Q_\tau - \gamma P_\theta Q_\theta). \tag{9}$$

In particular, the Q equation can first be written as $\partial_\tau(e^{2P}Q_\tau) - \partial_\theta(e^{2P}\gamma Q_\theta) = 0$. The stationary equations are $\mathcal{E}_P = \mathcal{E}_Q = 0$.

Define the densities

$$H_G = P_\tau^2 + \gamma P_\theta^2 + e^{2P}(Q_\tau^2 + \gamma Q_\theta^2),$$
$$M_G = 2(P_\tau P_\theta + e^{2P} Q_\tau Q_\theta). \tag{10}$$

The remaining reconstruction equations are $\lambda_\tau = H_G$ and $\lambda_\theta = M_G$.

Proposition 3.1 (Local integrability and global periodicity)**.** *For twice continuously differentiable P, Q,*

$$\partial_\tau M_G - \partial_\theta H_G = 2P_\theta \mathcal{E}_P + 2e^{2P} Q_\theta \mathcal{E}_Q. \tag{11}$$

For a solution on $I \times \mathbb{T}$, a single-valued periodic λ with these two derivatives exists if and only if $\int_0^{2\pi} M_G \, d\theta = 0$ at one, and hence every, time. It is unique up to an additive constant.

Proof. Differentiate the two densities. The terms with $P_\tau P_{\tau\theta}$ and $Q_\tau Q_{\tau\theta}$ cancel, leaving

$$\begin{aligned}\partial_\tau M_G - \partial_\theta H_G = {} & 2P_\theta(P_{\tau\tau} - \gamma P_{\theta\theta}) \\ & + 2e^{2P}[Q_\theta(Q_{\tau\tau} - \gamma Q_{\theta\theta}) + 2P_\tau Q_\tau Q_\theta - P_\theta Q_\tau^2 - \gamma P_\theta Q_\theta^2].\end{aligned}$$

Substitute (9) and combine the two occurrences of $\gamma P_\theta Q_\theta^2$. This gives (11). On a solution the identity vanishes, so periodicity gives $d \int M_G \, d\theta / d\tau = 0$.

If a periodic λ exists, integrating $\lambda_\theta = M_G$ forces the zero-period condition. Conversely, fix τ_0 and define on the lifted strip

$$\lambda(\tau, \theta) = \lambda_0 + \int_{\tau_0}^{\tau} H_G(s, 0) \, ds + \int_0^{\theta} M_G(\tau, \eta) \, d\eta.$$

Its θ derivative is M_G. Its τ derivative is H_G by the integrability identity and the fundamental theorem of calculus. The zero-period condition makes it periodic. The difference of two such functions has both derivatives zero, so is constant. □

This separates a local compatibility condition from a global topology condition. Satisfying the evolution equations makes the reconstruction one-form closed. A vanishing spatial period makes it the derivative of a periodic potential.

3.1 An exact polarized mode

Set $Q = 0$ and take an integer $n \geq 1$. Define the entire function

$$J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m (x^2/4)^m}{(m!)^2}.$$

The series and its differentiated series converge uniformly on compact sets. Coefficient comparison gives $x^2 J_0'' + x J_0' + x^2 J_0 = 0$. Thus

$$P(\tau, \theta) = A J_0(n e^{-\tau}) \cos(n\theta) \tag{12}$$

is an exact polarized solution. Indeed, for $x = ne^{-\tau}$, $\partial_\tau^2 J_0(x) = x^2 J_0''(x) + x J_0'(x) = -x^2 J_0(x)$, whereas $P_{\theta\theta} = -n^2 P$. Also $M_G = 2P_\tau P_\theta$ has zero spatial integral, so periodic λ can be reconstructed.

The Taylor channels can be separated without changing the physical mode. Put $y = x^2/4$, $\mathcal{B}(y) = J_0(2\sqrt{y})$, and for an integer $r \geq 2$ define

$$\mathcal{B}_{r,j}(y) = \sum_{q=0}^{\infty} \frac{(-1)^{rq+j} y^{rq+j}}{((rq+j)!)^2}, \qquad j = 0, \ldots, r-1.$$

Then $\mathcal{B} = \sum_j \mathcal{B}_{r,j}$. With $\vartheta = y\,d/dy$, the coefficient identity

$$\vartheta^2 \mathcal{B}_{r,j} + y\mathcal{B}_{r,j-1} = 0, \qquad j \pmod r,$$

follows from $m^2/(m!)^2 = 1/((m-1)!)^2$ for $m \geq 1$. Consequently the separated Taylor channels are coupled. Their sum satisfies the Bessel equation. This decomposition resolves series order and does not introduce additional independent gravitational degrees of freedom.

4 An exact finite Gowdy constraint projection

Let $h = 2\pi/N$ and use the periodic centered derivative

$$(D_h f)_j = \frac{f_{j+1} - f_{j-1}}{2h}, \qquad (\Delta_h f)_j = \frac{f_{j+1} - 2f_j + f_{j-1}}{h^2}.$$

Here $D_h^{\mathsf{T}} = -D_h$. With $U = \dot{P}$, $V = \dot{Q}$, the raw semidiscrete vector field $f_h(\tau, z)$ for $z = (P, Q, U, V)$ is

$$\begin{aligned} \dot{P} &= U, & \dot{Q} &= V, \\ \dot{U} &= \gamma\Delta_h P + e^{2P}[V^2 - \gamma(D_h Q)^2], & & \\ \dot{V} &= \gamma\Delta_h Q - 2[UV - \gamma(D_h P)(D_h Q)]. & & \end{aligned} \tag{13}$$

Products and exponentials in these formulas are componentwise. The finite global momentum is

$$C_h(z) = 2h \sum_j [U_j (D_h P)_j + e^{2P_j} V_j (D_h Q)_j]. \tag{14}$$

It is linear in the velocity variables when the configuration P, Q is fixed.

For the ordinary Euclidean gradients, skew-adjointness of D_h gives

$$\begin{aligned} \nabla_P C_h &= -2hD_h U + 4he^{2P} V(D_h Q), \\ \nabla_Q C_h &= -2hD_h(e^{2P} V), \\ g_U := \nabla_U C_h &= 2hD_h P, & g_V := \nabla_V C_h &= 2he^{2P} D_h Q. \end{aligned}$$

Put $G_h = (0, 0, g_U, g_V)$ and $d_h = \|g_U\|_2^2 + \|g_V\|_2^2$. On the open region $d_h > 0$, define

$$\Lambda_h = \frac{\nabla C_h \cdot f_h}{d_h}, \qquad \hat{f}_h = f_h - \Lambda_h G_h.$$

The chain rule proves the exact tangency identity

$$\nabla C_h \cdot \hat{f}_h = \nabla C_h \cdot f_h - \Lambda_h d_h = 0.$$

Every differentiable solution that remains in this region therefore keeps C_h constant.

Proposition 4.1 (Exact endpoint momentum)**.** *Start from z_n and let a provisional step produce $\tilde{z} = (\widetilde{P}, \widetilde{Q}, \widetilde{U}, \widetilde{V})$. Form $\tilde{g}_U, \tilde{g}_V$ at its configuration and assume $\tilde{d}_h > 0$. The correction*

$$\mu = \frac{C_h(\tilde{z}) - C_h(z_n)}{\tilde{d}_h}, \qquad z_{n+1} = (\widetilde{P}, \widetilde{Q}, \widetilde{U} - \mu\tilde{g}_U, \widetilde{V} - \mu\tilde{g}_V)$$

satisfies $C_h(z_{n+1}) = C_h(z_n)$ exactly. It is the correction of smallest Euclidean velocity norm at the fixed configuration.

Proof. Linearity in the velocities gives $C_h(z_{n+1}) = C_h(\tilde{z}) - \mu\tilde{d}_h$. For minimality, write $g = (\tilde{g}_U, \tilde{g}_V)$ and $b = C_h(z_n) - C_h(\tilde{z})$. Every admissible velocity displacement v satisfies $g \cdot v = b$. Cauchy–Schwarz implies $\|v\| \geq |b|/\|g\|$, with equality only at $v = bg/\|g\|^2$. This is the displayed correction. □

Proposition 4.2 (Fixed-mesh second-order time accuracy)**.** *Fix h. Suppose a smooth exact solution of $\dot{z} = \hat{f}_h(\tau, z)$ and its numerical steps remain in a compact regular neighborhood where $d_h \geq d_0 > 0$. Assume the mixed time and state derivatives of $\hat{f}_h$ through order three and the state derivatives of C_h through order four are bounded there. Apply the endpoint correction of Proposition 4.1 after explicit midpoint. Then its local error is $O(\Delta\tau^3)$ and its global error on a fixed time interval is $O(\Delta\tau^2)$.*

Proof. The explicit midpoint expansion agrees with the Taylor expansion of the exact flow through order two. Its provisional endpoint differs from the exact endpoint by $O(\Delta\tau^3)$. Since the exact flow keeps C_h constant and C_h has bounded first derivatives, the provisional constraint defect is also $O(\Delta\tau^3)$. The lower bound on the denominator and the upper bound on g make the correction $O(\Delta\tau^3)$. Thus the corrected local error has the same order. Under the stated derivative bounds the corrected step map has Lipschitz constant at most $1 + L\Delta\tau$ for small steps. If e_n is its error against the exact solution, then

$$e_{n+1} \leq (1 + L\Delta\tau)e_n + M\Delta\tau^3.$$

Iterating this inequality for $n\Delta\tau \leq T$ bounds the error by $MTe^{LT}\Delta\tau^2$ when $e_0 = 0$. □

These are regular-chart results. For a spatially homogeneous configuration, $g_U = g_V = 0$, and the displayed multiplier is undefined even though $C_h = 0$. Constants in the accuracy theorem can depend on h. A claim about simultaneous spatial and temporal refinement additionally needs bounds uniform in h.

The local compatibility diagnostic is a separate quantity. Define $H_{G,h}, M_{G,h}$ by replacing spatial derivatives in (10) with D_h. If $\mathcal{E}_{P,h} = \dot{U} - \dot{U}_{\text{raw}}$ and $\mathcal{E}_{Q,h} = \dot{V} - \dot{V}_{\text{raw}}$, then

$$\mathcal{J}_h = \partial_\tau M_{G,h} - D_h H_{G,h} - 2(D_h P)\mathcal{E}_{P,h} - 2e^{2P}(D_h Q)\mathcal{E}_{Q,h}$$

subtracts the explicit constraint-force contribution. Exact global momentum preservation does not force $\mathcal{J}_h$ to vanish at each site. Its refinement behavior tests whether the modified evolution approaches the local continuum reconstruction identity. The archived second-order numerical observations are experimental evidence for that behavior, with their tested range recorded in the companion.

5 Finite linearized ADM gravity

The symmetry-reduced model can now be compared with a finite three-dimensional linearized constraint system. Its exact algebraic closure rests on commuting linear derivatives. Nonlinear products will introduce an additional requirement in the next section.

5.1 Finite calculus and the quadratic action

Let V_h be a finite real inner-product space. Let $\delta_1, \delta_2, \delta_3$ be commuting skew-adjoint operators on it:

$$\delta_i\delta_j = \delta_j\delta_i, \qquad \langle \delta_i X, Y\rangle_h = -\langle X, \delta_i Y\rangle_h , \qquad \delta^2 = \sum_i \delta_i^2.$$

The periodic centered differences in three coordinate directions provide one example. All tensor inner products below sum over both indices, using the Frobenius inner product on symmetric tensors.

Let $h_{ij} = h_{ji}$ and $\pi_{ij} = \pi_{ji}$ take values in V_h. Repeated spatial indices are summed. Put $h_{\mathrm{tr}} = h_{kk}$, $\pi_{\mathrm{tr}} = \pi_{kk}$ and define

$$\begin{aligned} C &= \delta_i\delta_j h_{ij} - \delta^2 h_{\mathrm{tr}}, & C_i &= -2\delta_j \pi_{ij}, \\ \mathcal{G}_{ij}(h) &= \frac{1}{2}\big[\delta_k\delta_i h_{jk} + \delta_k\delta_j h_{ik} - \delta^2 h_{ij} - \delta_i\delta_j h_{\mathrm{tr}} - \delta_{ij} C\big]. \end{aligned} \tag{15}$$

Lemma 5.1 (Linearized finite Bianchi and adjoint identities)**.** *The spatial operator obeys $\delta_j \mathcal{G}_{ij}(h) = 0$ and is self-adjoint on symmetric tensors.*

Proof. Twice its divergence expands as

$$\begin{aligned} 2\delta_j\mathcal{G}_{ij} = {} & \delta_i\delta_j\delta_k h_{jk} + \delta^2\delta_k h_{ik} - \delta^2\delta_j h_{ij} - \delta_i\delta^2 h_{\mathrm{tr}} \\ & - \delta_i\delta_k\delta_l h_{kl} + \delta_i\delta^2 h_{\mathrm{tr}}. \end{aligned}$$

The terms cancel in three pairs. For self-adjointness, write $B_i(a) = \delta_j a_{ij}$ and $A(a) = \delta_i\delta_j a_{ij}$. Two applications of the adjoint identity give

$$\begin{aligned} 2\langle a, \mathcal{G}(b)\rangle_h = {} & -2\langle B(a), B(b)\rangle_h + \sum_k \langle \delta_k a, \delta_k b\rangle_h \\ & - \langle A(a), b_{\mathrm{tr}}\rangle_h - \langle a_{\mathrm{tr}}, A(b)\rangle_h - \sum_k \langle \delta_k a_{\mathrm{tr}}, \delta_k b_{\mathrm{tr}}\rangle_h . \end{aligned}$$

This expression is symmetric in a, b. □

Set

$$T_h = \langle \pi, \pi\rangle_h - \tfrac{1}{2}\langle \pi_{\mathrm{tr}}, \pi_{\mathrm{tr}}\rangle_h , \qquad U_h = \tfrac{1}{2}\langle h, \mathcal{G}(h)\rangle_h .$$

With lapse perturbation n and shift perturbation n_i, the action is

$$S_h^{(2)} = \int \Big[\big\langle \pi, \dot{h}\big\rangle_h - T_h - U_h + \langle n, C\rangle_h - \langle n_i, C_i\rangle_h\Big]\, dt. \tag{16}$$

Variation of n, n_i imposes $C = C_i = 0$. The remaining equations are

$$\begin{aligned} \dot{h}_{ij} &= 2\pi_{ij} - \delta_{ij}\pi_{\mathrm{tr}} + \delta_i n_j + \delta_j n_i, \\ \dot{\pi}_{ij} &= -\mathcal{G}_{ij}(h) + \delta_i\delta_j n - \delta_{ij}\delta^2 n. \end{aligned} \tag{17}$$

The kinetic gradient is $2\pi_{ij} - \delta_{ij}\pi_{\mathrm{tr}}$. Lemma 5.1 makes the potential gradient $\mathcal{G}(h)$. The lapse gradient follows by two spatial integrations by parts. Integrating the shift term once gives $-\langle \delta_i n_j + \delta_j n_i, \pi_{ij}\rangle_h$ in the Lagrangian, fixing the sign in the first equation.

5.2 Constraint propagation and the canonical brackets

Theorem 5.2 (Exact finite linearized closure)**.** *For every solution of* (17)*,*

$$\dot{C} = -\delta_i C_i, \qquad \dot{C}_i = 0.$$

Thus initially vanishing constraints remain zero. With the canonical Frobenius bracket and smearings $C[n] = \langle n, C\rangle_h$*,* $D[\xi] = \langle \xi_i, C_i\rangle_h$*, all three brackets* $\{C[n], C[m]\}$*,* $\{D[\xi], D[\eta]\}$*, and* $\{C[n], D[\xi]\}$ *vanish identically.*

Proof. The trace of the momentum part of $\dot{h}$ is $-\pi_{\mathrm{tr}}$. Its contribution to $\dot{C}$ is therefore $2\delta_i\delta_j\pi_{ij} = -\delta_i C_i$. The shift contribution is

$$\delta_i\delta_j(\delta_i n_j + \delta_j n_i) - \delta^2(2\delta_k n_k) = 0.$$

For $\dot{C}_i$, the divergence of $\mathcal{G}$ vanishes by Lemma 5.1, and $\delta_j(\delta_i\delta_j n - \delta_{ij}\delta^2 n) = 0$. This proves the propagation equations and their stated implication.

For the bracket, use $\{F, G\} = \langle \nabla_h F, \nabla_\pi G\rangle_h - \langle \nabla_\pi F, \nabla_h G\rangle_h$. Two scalar constraints depend only on h and two momentum constraints depend only on π, so their brackets vanish. The gradients for the mixed bracket are

$$(\nabla_h C[n])_{ij} = \delta_i\delta_j n - \delta_{ij}\delta^2 n, \qquad (\nabla_\pi D[\xi])_{ij} = \delta_i\xi_j + \delta_j\xi_i.$$

Their contraction vanishes after one integration by parts because the first tensor is divergence-free. □

The momentum generator changes the metric perturbation by $h_{ij} \mapsto h_{ij} + \delta_i\xi_j + \delta_j\xi_i$. The corresponding change of the scalar constraint is $2\delta^2\delta_k\xi_k - 2\delta^2\delta_k\xi_k = 0$. No product rule is used in these cancellations.

For a partition of fine spatial sites into coarse regions A, the nonnegative diagnostic

$$E_A^{\mathrm{con}} = \sum_{x\in A} h^3 \left(|C_x|^2 + \sum_i |C_{i,x}|^2 \right)$$

adds exactly to the full fine diagnostic. This is an exact partition identity. It does not assert that differentiating a coarse field reconstructs its fine constraint values.

5.3 Tensor waves, positive reduced energy, and resolved dispersion

Choose $n = n_i = 0$ and impose the transverse and traceless conditions

$$\delta_j h_{ij} = 0, \quad h_{\mathrm{tr}} = 0, \qquad \delta_j\pi_{ij} = 0, \quad \pi_{\mathrm{tr}} = 0.$$

They are preserved by the evolution because

$$\mathcal{G}_{ij}(h) = -\tfrac{1}{2}\delta^2 h_{ij}, \qquad \dot{h} = 2\pi, \qquad \dot{\pi} = \tfrac{1}{2}\delta^2 h, \qquad \ddot{h} = \delta^2 h.$$

On this reduced sector the energy is

$$H_{\mathrm{TT}} = \|\pi\|_h^2 + \frac{1}{4}\sum_k \|\delta_k h\|_h^2 \geq 0.$$

Its derivative is zero: substituting the two first-order equations reduces it to $\langle \pi, \delta^2 h\rangle_h + \sum_k \langle \delta_k h, \delta_k \pi\rangle_h = 0$.

For a Fourier mode on the periodic centered lattice, let $\tilde{k}_i = \sin(k_i h)/h$. If $\tilde{\mathbf{k}} \neq 0$, define

$$\Pi_{ij} = \delta_{ij} - \frac{\tilde{k}_i \tilde{k}_j}{|\tilde{\mathbf{k}}|^2}, \qquad \Lambda^{\mathrm{TT}}_{ij,kl} = \tfrac{1}{2}(\Pi_{ik}\Pi_{jl} + \Pi_{il}\Pi_{jk} - \Pi_{ij}\Pi_{kl}).$$

For a symmetric tensor X, its image is $\Pi X \Pi - \Pi \operatorname{tr}(\Pi X)/2$. It is transverse and traceless because $\Pi^2 = \Pi$ and $\operatorname{tr}\Pi = 2$. Applying the map again leaves it unchanged. On the transverse two-plane, symmetric tensors have dimension three and the trace condition removes one dimension. Thus the projector has rank two.

The resulting frequency obeys

$$\omega_h^2 = \sum_i \frac{\sin^2(k_i h)}{h^2}. \tag{18}$$

For fixed $\mathbf{k} \neq 0$ as $h \to 0$,

$$\omega_h^2 = |\mathbf{k}|^2 - \frac{h^2}{3}\sum_i k_i^4 + O(h^4), \qquad \omega_h = |\mathbf{k}| - \frac{h^2}{6|\mathbf{k}|}\sum_i k_i^4 + O(h^4).$$

These follow by expanding the sine and then the positive square root. The regime keeps $\mathbf{k}$ fixed. It is not uniform up to the Nyquist boundary.

The spatial operator in (18) is the sum of the squared centered first derivatives. It differs from the nearest-neighbor second-difference operator used for the raw Gowdy equations. For even site counts, an alternating sequence has centered derivative zero. Such checkerboard modes, as well as constant modes, belong to the derivative kernel. At a zero symbol the above TT projector is undefined and the rank-two argument does not apply. Exact constraint propagation consequently coexists with a finite spatial kernel that a physical continuum comparison must address.

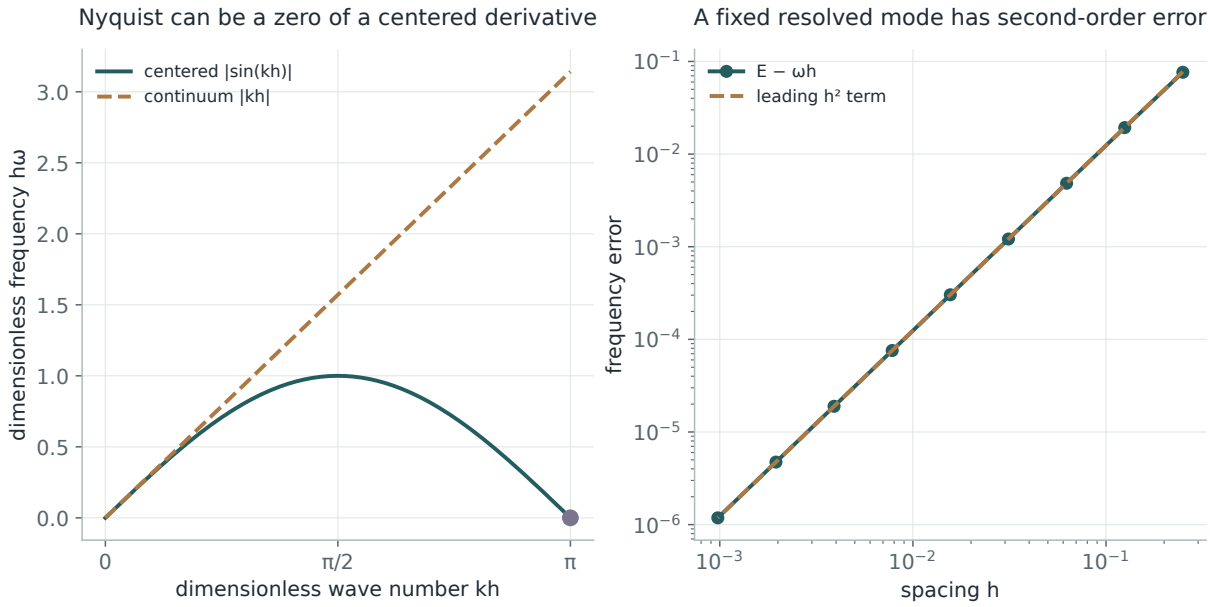


Figure 2. The centered derivative has a Nyquist zero, while a fixed resolved mode approaches its continuum frequency with second-order error. The left panel is the one-dimensional symbol; the right panel tests a fixed mode under refinement.

6 The nonlinear pointwise obstruction and its algebraic alternative

The linearized cancellations use commuting derivatives but never ask them to differentiate a product. A nonlinear diffeomorphism generator does. The following scalar model isolates that issue without relying on the more involved gravitational tensor constraints.

Let a periodic real lattice have canonical brackets $\{q_a, p_b\}_h = \delta_{ab}/h$ and a skew matrix D. Define

$$\mathcal{D}_h[\xi] = h\sum_a \xi_a p_a (Dq)_a, \qquad \mathfrak{L}_D(f,g) = D(fg) - (Df)g - f(Dg).$$

All products on this lattice are pointwise.

Proposition 6.1 (Exact nonlinear defect identity)**.** *For arbitrary finite arrays q, p, ξ, η,*

$$\begin{aligned}\{\mathcal{D}_h[\xi], \mathcal{D}_h[\eta]\}_h &= \mathcal{D}_h[\xi D\eta - \eta D\xi] \\ &\quad + h\sum_a p_a[\xi_a \mathfrak{L}_D(\eta, Dq)_a - \eta_a \mathfrak{L}_D(\xi, Dq)_a].\end{aligned} \tag{19}$$

Proof. Coordinate differentiation gives $\partial_p \mathcal{D}_h[\xi] = h\xi Dq$ and $\partial_q \mathcal{D}_h[\xi] = -hD(\xi p)$. Insert these in the canonical bracket and use $D^{\mathsf{T}} = -D$:

$$\{\mathcal{D}_h[\xi], \mathcal{D}_h[\eta]\}_h = h\sum_a p_a\left[\xi_a D(\eta Dq)_a - \eta_a D(\xi Dq)_a\right].$$

Expanding each derivative using the definition of $\mathfrak{L}_D$ cancels the two $\xi\eta D^2 q$ terms. The remaining product-rule terms give the first term of (19), and the two defect terms give the second. □

Theorem 6.2 (Finite pointwise derivation obstruction)**.** *A linear derivation on the complete pointwise algebra $\mathbb{C}^N$ is zero. Thus a nonzero finite pointwise derivative cannot obey the undeformed Leibniz rule on every pair of arrays.*

Proof. Let e_a be the coordinate idempotents. If D is a derivation, then $D(e_a) = D(e_a^2) = 2e_a D(e_a)$. At coordinate a this forces the component to be twice itself, hence zero. At every other coordinate the right side is zero. Therefore $D(e_a) = 0$ for every a. Linearity and the coordinate basis imply $D = 0$. □

The theorem rules out a universal ordinary derivation on the full finite pointwise algebra. For a particular generator, one must still inspect the contracted defect in (19). A nonzero Leibniz defect does not by itself imply that every contraction is nonzero.

Example 6.3 (An exact bracket failure)**.** Use $N = 5$, $h = 1$, and $(Df)_a = (f_{a+1} - f_{a-1})/2$. Take $q = e_0$, $p = \xi = e_1$, and $\eta = e_2$. Then $(Dq)_1 = -1/2$, $(D\eta)_1 = 1/2$, and $\eta Dq = 0$. The direct bracket in the preceding proof is zero, while

$$\mathcal{D}_h[\xi D\eta - \eta D\xi] = -\tfrac{1}{4}.$$

The defect is therefore exactly $+1/4$. This finite witness shows that the naive centered scalar constraint does fail to close universally in the proposed undeformed form.

6.1 Smooth resolved arrays and the matrix corner route

For samples of sufficiently smooth periodic functions, the centered derivative obeys

$$D_h f = f' + \frac{h^2}{6} f''' + O(h^4).$$

If f, g have uniformly bounded derivatives through order five, its Leibniz defect consequently satisfies

$$\mathfrak{L}_{D_h}(f, g) = \frac{h^2}{2}(f''g' + f'g'') + O(h^4). \tag{20}$$

For fixed smooth p, ξ, η, q, with sufficiently many additional bounded derivatives of q to control $D_h q$, the contraction in (19) is $O(h^2)$ on a fixed periodic interval. Each summand is uniformly $O(h^2)$ and hN is the interval length. The coordinate-idempotent witness above is outside this uniformly smooth regime.

A spectral derivative has an exact product rule on trigonometric polynomials when every product remains in the represented band without aliasing. Repeated products require further headroom. These restricted domains differ from the complete pointwise algebra. The later resolved constraint theorem includes explicit domain bounds.

For the full matrix algebra, choose a Hermitian matrix K and set $\delta_K X = i[K, X]$. Direct multiplication gives

$$\delta_K(XY) = (\delta_K X)Y + X(\delta_K Y). \tag{21}$$

It also obeys $\delta_K(X^*) = (\delta_K X)^*$, $\operatorname{Tr} \delta_K X = 0$, and is skew-adjoint for the Hilbert–Schmidt inner product. The last identity follows by moving K through the trace. These properties supply an exact finite product calculus on a noncommutative algebra.

Nonlinear algebra and regular reduction

7 Nonlinear closure and the return of discarded detail

The nonlinear construction has two distinct tasks. A fine theory must first have a consistent gauge differential. Its passage to a retained resolution must then preserve that differential. The second task can be solved conditionally without solving the first. This distinction explains the sequence of the chapters: an obstruction to direct lattice closure is followed by a transfer mechanism, a resolved vacuum example, a real Lorentzian parent, and a local canonical reduction. Each step changes the mathematical setting. An earlier obstruction is not a refutation of a later construction on a different domain, and the later construction does not make the earlier hypotheses disappear.

7.1 An exact matrix momentum sector

Let the fine canonical coordinates be Hermitian matrix pairs (Z^A, P_A), with real components and the trace bracket

$$\{F, G\} = \sum_A \mathrm{Tr}\left(\nabla_{Z^A} F \nabla_{P_A} G - \nabla_{P_A} F \nabla_{Z^A} G\right). \tag{22}$$

The trace gradients are defined by real variations. For a Hermitian parameter Λ, put

$$D[\Lambda] = i \sum_A \mathrm{Tr}(P_A[\Lambda, Z^A]). \tag{23}$$

This real functional generates simultaneous conjugation:

$$\{Z^A, D[\Lambda]\} = i[\Lambda, Z^A], \qquad \{P_A, D[\Lambda]\} = i[\Lambda, P_A]. \tag{24}$$

Indeed, cyclic variation gives $\nabla_P D[\Lambda] = i[\Lambda, Z]$ and $\nabla_Z D[\Lambda] = i[P, \Lambda]$ for one pair.

Proposition 7.1 (Finite inner momentum closure)**.** *With these conventions,*

$$\{D[\Lambda], D[\Gamma]\} = D[i[\Lambda, \Gamma]]. \tag{25}$$

If a matrix density $\mathcal{E}$ obeys $\{\mathcal{E}, D[\Lambda]\} = i[\Lambda, \mathcal{E}]$, then $H[N] = \mathrm{Tr}(N\mathcal{E})$ satisfies

$$\{D[\Lambda], H[N]\} = H[i[\Lambda, N]]. \tag{26}$$

Proof. For one pair, substitution of the two cyclic gradients yields

$$\begin{aligned} \{D[\Lambda], D[\Gamma]\} &= \mathrm{Tr}\left(-[P, \Lambda][\Gamma, Z] + [\Lambda, Z][P, \Gamma]\right) && (27)\\ &= \mathrm{Tr}\left(P(-[\Lambda, [\Gamma, Z]] + [\Gamma, [\Lambda, Z]])\right) && (28)\\ &= -\mathrm{Tr}(P[[\Lambda, \Gamma], Z]) = D[i[\Lambda, \Gamma]]. && (29) \end{aligned}$$

Summation handles all pairs. For the second statement, antisymmetry and cyclicity give

$$\{D[\Lambda], H[N]\} = -i\,\mathrm{Tr}(N[\Lambda, \mathcal{E}]) = \mathrm{Tr}(i[\Lambda, N]\mathcal{E}). \tag{30}$$

□

The closed finite object is the full inner algebra. To identify a continuum spatial diffeomorphism, one must additionally construct parameters $\Lambda_h[\xi]$ whose action approximates the Lie derivative on resolved observables. An arbitrary subset of matrices chosen to resemble first derivatives need not be closed under commutators. Equation (25) does not establish that extra geometric identification.

7.2 What direct projection loses

Let $(\mathcal{C}_F, s_F)$ be a graded fine complex with $s_F^2 = 0$. Let $P : \mathcal{C}_F \to \mathcal{C}_C$ and $I : \mathcal{C}_C \to \mathcal{C}_F$ be degree-zero maps satisfying $PI = \mathrm{id}_C$, and set

$$Q = \mathrm{id}_F - IP, \qquad \bar{s}_C = Ps_F I. \tag{31}$$

Here I is the inclusion map, whereas id denotes an identity operator. Insertion of $IP = \mathrm{id}_F - Q$ proves the exact identity

$$\bar{s}_C^2 = Ps_F(IP)s_F I = -Ps_F Q s_F I. \tag{32}$$

The lost path starts in retained data, enters a discarded channel, and returns to retained data. Its sign and order are fixed by the formula. There is no reason for it to vanish merely because the fine differential is nilpotent.

For example, the finite transfer test has the naive coarse block

$$\bar{s}_C = \begin{pmatrix} -1/25 & -1/25 & -26/625 \\ 1 & 0 & 1/25 \\ 0 & 1/25 & 0 \end{pmatrix}. \tag{33}$$

Its square is nonzero, as its $(2,2)$ entry is $-1/25$. The transferred block in that same test is

$$s_C = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1/25 \\ 0 & 0 & 0 \end{pmatrix}, \qquad s_C^2 = 0. \tag{34}$$

This small rational example illustrates the distinction between a projection and a transfer. The matrices by themselves are an algebraic diagnostic. They do not define gravitational constraints, a symplectic BFV theory, or a continuum model.

7.3 A precise conditional transfer theorem

Split $s_F = s_0 + \delta$, where $s_0^2 = 0$ and δ has degree one. Suppose $(\mathcal{C}_C, s_{0,C})$ is a coarse complex and the maps are chain maps:

$$s_0 I = I s_{0,C}, \qquad P s_0 = s_{0,C} P. \tag{35}$$

Assume a degree-minus-one homotopy H satisfies

$$\begin{aligned} PI = \mathrm{id}_C, \qquad & \mathrm{id}_F - IP = s_0 H + H s_0, \\ PH = 0, \qquad HI = 0, \qquad & H^2 = 0. \end{aligned} \tag{36}$$

Finally, require $(s_0+\delta)^2=0$ and an inverse of $\mathrm{id}_F+H\delta$. The inverse may exist in a declared operator topology, or in a complete filtration in which the Neumann series makes formal sense. A formal inverse is a formal conclusion, not an analytic convergence theorem.

Theorem 7.2 (Transfer of a differential)**.** *Under these assumptions, define*

$$I_\delta=(\mathrm{id}_F+H\delta)^{-1}I,\qquad s_C=s_{0,C}+P\delta I_\delta. \tag{37}$$

Then $PI_\delta=\mathrm{id}_C$, $(s_0+\delta)I_\delta=I_\delta s_C$, *and* $s_C^2=0$.

Proof. The inverse equation is $I_\delta=I-H\delta I_\delta$. Application of P proves $PI_\delta=PI=\mathrm{id}_C$. Let

$$K=(s_0+\delta)I_\delta-I_\delta s_C. \tag{38}$$

Using the chain-map identity and $\mathrm{id}_F-IP=s_0H+Hs_0$ in this expression gives

$$K=H\left(s_0\delta I_\delta+\delta I_\delta s_{0,C}+\delta I_\delta P\delta I_\delta\right). \tag{39}$$

Nilpotency of the fine differential gives $s_0\delta+\delta s_0+\delta^2=0$. The parenthesis therefore equals $-\delta K$. Hence $(\mathrm{id}_F+H\delta)K=0$, and invertibility gives $K=0$. Applying the fine differential to the intertwining identity yields $0=I_\delta s_C^2$. The left inverse P then gives $s_C^2=0$. □

Expansion of the inverse exhibits the returned detail:

$$\begin{aligned} s_C=s_{0,C}&+P\delta I-P\delta H\delta I\\ &+P\delta H\delta H\delta I-\cdots. \end{aligned} \tag{40}$$

The first correction describes one propagation through H between two nonlinear vertices. More interactions require further terms. Truncating this series generally changes the nilpotency statement.

There is a separate step from this theorem to a nonlinear BFV charge. On a canonical graded phase space, introduce odd ghosts c^A of ghost number one and their odd conjugate momenta ϖ_A of ghost number minus one. A charge of ghost number one has the form

$$\Omega_F=c^AC_A-\tfrac{1}{2}c^Ac^B{U_{AB}}^C\varpi_C+\Omega_F^{(\geq 3)}. \tag{41}$$

The superscript counts ghost factors, not total ghost number. Higher terms can have further ghost momenta and still have ghost number one. They carry the additional identities needed for field-dependent, open, or reducible algebras. They cannot be suppressed just because the displayed constraint brackets close.

If $\{\Omega_F,\Omega_F\}=0$, then $s_F=\{\Omega_F,\cdot\}$ is a differential. To transfer a nonlinear Hamiltonian gauge structure, the contraction must also lift compatibly to the completed nonlinear coalgebra and respect its cyclic or symplectic structure. On a local exact canonical chart, a resulting Hamiltonian vector field can have a charge Ω_C. Its square then has Hamiltonian $\frac{1}{2}\{\Omega_C,\Omega_C\}$, which vanishes when the canonical center has no ghost-number-two constants. These are additional assumptions. Connectedness alone does not turn every symplectic vector field into a global Hamiltonian vector field.

7.4 A quantitative return bound

For bounded maps on declared normed spaces, assume

$$\|H\delta\| \le \rho < 1, \qquad \|Q\delta I\| \le Ch^r. \tag{42}$$

The resolvent identity and $H = HQ$ give

$$s_C - (s_{0,C} + P\delta I) = -P\delta(\mathrm{id}_F + H\delta)^{-1} H\delta I, \tag{43}$$

$$\|s_C - (s_{0,C} + P\delta I)\| \le \frac{\|P\delta\|\,\|H\|}{1-\rho}\, Ch^r. \tag{44}$$

To make this an approximation theorem along a refinement sequence, the prefactor must be bounded uniformly in h. An unbounded detail Green operator can defeat the conclusion even when $Q\delta I$ is small. The estimate controls the stated differential maps. It does not by itself prove convergence of constrained trajectories, physical observables, or a quantum measure.

8 A resolved nonlinear vacuum algebra

8.1 A filtered algebra with room for products

On $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, let

$$\mathcal{T}_K = \left\{ \sum_{\|k\|_\infty \le K} \hat{f}_k e^{ik\cdot x} \right\}. \tag{45}$$

The algebraic relations are

$$\mathcal{T}_K \mathcal{T}_L \subset \mathcal{T}_{K+L}, \qquad \partial_a \mathcal{T}_K \subset \mathcal{T}_K. \tag{46}$$

Thus $\mathcal{T}_K$ is a resolved space, while $\bigcup_K \mathcal{T}_K$ is the product-closed differential algebra. Let S_N sample on an N^3 periodic grid, δ_a be its Fourier derivative, and $\tau_N = N^{-3}\sum_x$ its normalized trace.

If $f_j \in \mathcal{T}_{K_j}$ and $\sum_j K_j < N/2$, the product stays strictly below Nyquist in each coordinate. Sampling is injective on this support and gives

$$\delta_a \prod_j S_N f_j = S_N \partial_a \left(\prod_j f_j \right), \tag{47}$$

$$\tau_N \left(\prod_j S_N f_j \right) = (2\pi)^{-3} \int_{\mathbb{T}^3} \prod_j f_j \, \mathrm{d}^3x. \tag{48}$$

The Leibniz rule and trace integration by parts consequently hold for the declared products. They do not hold for arbitrary sampled pointwise products. This is how the resolved construction avoids the finite pointwise derivation obstruction: it retains enough detail instead of demanding a derivation on the whole finite function algebra.

8.2 Constraints and their normalizations

Take connection coefficients A_a^i, densitized triads E_i^a, internal parameters λ^i, shifts ξ^a, and densitized lapse n in $\mathcal{T}_K$. Set the canonical normalization to one and define

$$F_{ab}^i = \delta_a A_b^i - \delta_b A_a^i + \epsilon^i{}_{jk} A_a^j A_b^k, \tag{49}$$

$$\mathcal{G}_i = \delta_a E_i^a + \epsilon_{ij}{}^k A_a^j E_k^a, \tag{50}$$

$$\mathcal{V}_a = F_{ab}^i E_i^b, \qquad \mathcal{D}_a = \mathcal{V}_a - A_a^i \mathcal{G}_i, \tag{51}$$

$$\mathcal{H} = \tfrac{1}{2}\epsilon^{ij}{}_k E_i^a E_j^b F_{ab}^k + \Lambda \det E. \tag{52}$$

Here $\det E = \frac{1}{6}\epsilon_{abc}\epsilon^{ijk} E_i^a E_j^b E_k^c$. The smeared generators are

$$G[\lambda] = \tau_N(\lambda^i \mathcal{G}_i), \qquad D[\xi] = \tau_N(\xi^a \mathcal{D}_a), \tag{53}$$

$$V[\xi] = \tau_N(\xi^a \mathcal{V}_a), \qquad H[n] = \tau_N(n\mathcal{H}). \tag{54}$$

On the full grid canonical phase space the bracket is

$$\{F, G\}_N = N^3 \sum_{x,a,i} \left(\frac{\partial F}{\partial A_a^i(x)} \frac{\partial G}{\partial E_i^a(x)} - \frac{\partial F}{\partial E_i^a(x)} \frac{\partial G}{\partial A_a^i(x)} \right). \tag{55}$$

The factor N^3 matches the normalized trace in the action. The resolved identities below are evaluated at band-limited data in this phase space. Replacing this bracket by a bracket that first projects every gradient back to $\mathcal{T}_K$ is a different operation and can reintroduce the leakage of equation (32).

Put

$$(\lambda \times \mu)^i = \epsilon^i{}_{jk} \lambda^j \mu^k, \tag{56}$$

$$[\xi, \eta]^a = \xi^b \delta_b \eta^a - \eta^b \delta_b \xi^a, \tag{57}$$

$$\mathcal{L}_\xi^{(-1)} n = \xi^a \delta_a n - n \delta_a \xi^a, \tag{58}$$

$$S^a(n, m) = E_i^a E^{bi} (m \delta_b n - n \delta_b m). \tag{59}$$

The lapse convention fixes the sign and density weight of this last expression. It should not be compared with a differently ordered normal bracket without translating conventions.

Proposition 8.1 (The resolved algebraic transfer step)**.** *Assume the continuum polynomial constraint identities in this convention and* $16K < N$*. Their full-grid sampled evaluations obey*

$$\{G[\lambda], G[\mu]\}_N = G[\lambda \times \mu], \tag{60}$$

$$\{D[\xi], G[\lambda]\}_N = G[\xi^a \delta_a \lambda], \tag{61}$$

$$\{D[\xi], D[\eta]\}_N = D[[\xi, \eta]], \tag{62}$$

$$\{G[\lambda], H[n]\}_N = 0, \tag{63}$$

$$\{D[\xi], H[n]\}_N = H[\mathcal{L}_\xi^{(-1)} n], \tag{64}$$

$$\{H[n], H[m]\}_N = V[S(n, m)] = D[S(n, m)] + G[S^a(n, m) A_a]. \tag{65}$$

Proof. Including the smearing, the Gauss, improved momentum, and Hamiltonian functionals contain at most three, four, and five band-K factors. Their canonical gradients contain at most two, three, and four. A bracket therefore contains at most eight factors. The hypothesis gives $8K < N/2$. Equation (48)

identifies every product, derivative, integration by parts, and zero-mode extraction in the continuum derivation with its sampled value. The continuum identities transfer term by term. The cosmological term is gauge and diffeomorphism invariant. Its cross terms in the Hamiltonian bracket cancel by antisymmetry. The identity $V[S] = D[S] + G[S^a A_a]$ follows directly from the definition of $\mathcal{D}_a$. □

The proposition isolates the spectral transfer argument. The continuum connection calculation is its stated input. Neither a floating-point test nor a source reference can replace that input. The construction is an ordinary mathematical result rather than a certified Lean theorem.

8.3 What the closure tower does and does not preserve

At a resolved point, a Hamiltonian vector field can immediately generate higher bands. In particular, the fixed space $\mathcal{T}_K$ is not generally invariant under nonlinear evolution. The valid procedure enlarges the representation before the newly generated support aliases. Fourier zero padding satisfies

$$I_{L\to M} I_{K\to L} = I_{K\to M}. \tag{66}$$

When the larger band and its grid resolve every generated product, the embedded constraints and their bracket values agree. This is compatibility of finite algebraic evaluations. A tower of such evaluations does not yet supply a uniform interval of existence, a bound on band growth, or a continuum evolution theorem.

The source certificate tests the algebra at $N = 18$, $K = 1$, where $16K < N$. Its bracket residuals are of order 10^{-17}. A band-three negative control violates the hypothesis and produces a Hamiltonian-bracket residual near 3.3×10^{-8}. These are useful diagnostics of the implementation and of the support boundary. A finite test is not a proof for all admissible fields.

The Lorentzian limitation of this self-dual chart is also explicit. A single Lorentzian self-dual connection is complex, with a physical sector requiring real E and $A + \overline{A} = 2\Gamma(E)$. The source chapter does not construct a finite step preserving that sector. The real parent below addresses reality by changing variables and retaining both chiral sectors, rather than by declaring this missing step solved inside the single complex chart.

8.4 The separate Gowdy BFV target

The Gowdy target chapter changes from τ to areal time $t = e^{-\tau} > 0$. Its wave-map Hamiltonian is

$$\mathcal{H}_t = \frac{\Pi_P^2 + e^{-2P}\Pi_Q^2}{2t} + \frac{t}{2}(P_\theta^2 + e^{2P}Q_\theta^2). \tag{67}$$

The physical Hamiltonian density at fixed t and the tangential density $\mathcal{D} = \Pi_P P_\theta + \Pi_Q Q_\theta$ have the continuum bracket pattern of two deformation generators with densitized smearings. This algebraic target does not make $\mathcal{H}_t$ a vanishing gravitational constraint. Areal time is already a gauge choice.

A gravitational BFV construction must begin with the ungauge-fixed reduced Einstein action, retain its clock or embedding variables, derive its constraints, and recover the displayed physical Hamiltonian after gauge fixing. The source lists fine nilpotency, finite transfer, refinement consistency, continuum recovery, and quantum measure control as successive gates. The resolved vacuum example supplies a different classical input and does not close those Gowdy-specific gates by substitution. In particular, second-order defects of a direct difference approximation to the target bracket are not a finite BFV nilpotency theorem.

9 A real Lorentzian parent and its classical master equation

9.1 Fields, action, and polynomial structure

Fix $\eta_{IJ} = \mathrm{diag}(-1,1,1,1)$ and $\epsilon_{0123} = 1$. The independent fields are a real coframe e^I, a real antisymmetric Lorentz connection ω^{IJ}, an Abelian connection a, an antisymmetric auxiliary scalar φ_{IJ}, and Dirac fields $\psi, \bar{\psi}$. The spinors are Grassmann odd. Define

$$R^{IJ} = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \qquad F = \mathrm{d}a, \tag{68}$$

$$\mathrm{vol}_e = \tfrac{1}{4!}\epsilon_{IJKL} e^I \wedge e^J \wedge e^K \wedge e^L, \qquad \Sigma_I = \tfrac{1}{3!}\epsilon_{IJKL} e^J \wedge e^K \wedge e^L. \tag{69}$$

Use $\{\gamma^I, \gamma^J\} = 2\eta^{IJ}$ and $\gamma_{IJ} = \frac{1}{2}[\gamma_I, \gamma_J]$. The covariant derivatives are

$$\mathcal{D}\psi = \mathrm{d}\psi + \tfrac{1}{4}\omega^{IJ}\gamma_{IJ}\psi - iqa\psi, \tag{70}$$

$$\mathcal{D}\bar{\psi} = \mathrm{d}\bar{\psi} - \tfrac{1}{4}\bar{\psi}\omega^{IJ}\gamma_{IJ} + iqa\bar{\psi}. \tag{71}$$

The parameter q here is electric charge. The determinant of a spatial metric below will be written $\det(q_{ab})$ to avoid confusing the two uses.

The parent action is $S_0 = S_{\mathrm{G}} + S_{\mathrm{M}}^{(1)} + S_{\mathrm{D}}$, with

$$S_{\mathrm{G}} = \frac{1}{4\kappa}\int \epsilon_{IJKL} e^I \wedge e^J \wedge R^{KL} - \frac{\Lambda}{\kappa}\int \mathrm{vol}_e, \tag{72}$$

$$S_{\mathrm{M}}^{(1)} = \int \left(\tfrac{1}{2}\varphi_{IJ} F \wedge e^I \wedge e^J - \tfrac{1}{4}\varphi_{IJ}\varphi^{IJ}\,\mathrm{vol}_e\right), \tag{73}$$

$$S_{\mathrm{D}} = \frac{i}{2}\int \Sigma_I \wedge \left(\bar{\psi}\gamma^I \mathcal{D}\psi - (\mathcal{D}\bar{\psi})\gamma^I\psi\right) - m\int \mathrm{vol}_e\, \bar{\psi}\psi. \tag{74}$$

This formulation uses no inverse tetrad in its defining action. Its largest monomials have six field factors. Nondegeneracy becomes necessary when the auxiliary variables are eliminated or a metric interpretation is imposed.

The geometric fields are real, but the physical spinor convention needs a separate check. The displayed action retains the inherited Dirac-type polynomial gauge model. With the stated Clifford sign and kinetic factor i, the flat-space operators $i\gamma^I\partial_I \pm m$ have product

$$(i\gamma^I\partial_I + m)(i\gamma^J\partial_J - m) = -\Box - m^2, \qquad \Box = \eta^{IJ}\partial_I\partial_J. \tag{75}$$

Thus its squared field equation is $\Box + m^2 = 0$, whereas the conventional real massive dispersion in this mostly-plus metric uses $\Box - m^2 = 0$. A consistent translation of the Clifford, kinetic, mass, and adjoint conventions is required before identifying this sector with conventional physical massive Dirac matter. Gauge invariance and the graded master-equation argument do not require that physical identification, and they do not resolve it. The massless algebraic sector does not have this squared-mass discrepancy.

Variation of φ gives the unambiguous wedge equation

$$F \wedge e^I \wedge e^J = \varphi^{IJ}\,\mathrm{vol}_e\,. \tag{76}$$

Under the standard Lorentzian Hodge convention $\alpha \wedge \star\beta = \langle\alpha,\beta\rangle\,\mathrm{vol}_e$, this equation identifies φ with minus the internal components of $\star F$. For example, $F = Ee^0 \wedge e^1$ has $\star F = -Ee^2 \wedge e^3$, whereas the

wedge equation gives $\varphi^{23} = E$. The inherited shorthand $\varphi = \star F$ needs this sign translation. Substitution of the wedge equation into the action gives the conventional result

$$S_{\mathrm{M}}^{(1)}\big|_{\varphi} = -\tfrac{1}{4}\int F_{IJ}F^{IJ}\,\mathrm{vol}_e\,. \tag{77}$$

The substitution repairs the Hodge shorthand while retaining the on-shell Maxwell action.

9.2 Gauge transformations and graded conventions

For even parameters (ξ, λ, u), the coframe transforms by its Lie derivative and Lorentz rotation, the connection by $\mathcal{L}_\xi\omega - D_\omega\lambda$, and the Abelian potential by $\mathcal{L}_\xi a + \mathrm{d}u$. The spinor transforms by $\mathcal{L}_\xi\psi + \frac{1}{4}\lambda^{IJ}\gamma_{IJ}\psi + iqu\psi$, with the dual action on $\bar\psi$. All internal contractions in (74) are invariant. A diffeomorphism changes its four-form density by a Lie derivative, which integrates to zero on a periodic domain or under stated boundary conditions.

The parameter algebra combines vector fields with Lorentz and Abelian transformations in a semidirect product. In the inherited field-space commutator convention,

$$[\delta_U, \delta_V]_{\mathrm{fld}} = -\delta_{[U,V]_{\mathrm{par}}}, \tag{78}$$

where

$$[(\xi,\lambda,u),(\eta,\mu,v)]_{\mathrm{par}} = \big([\xi,\eta], \mathcal{L}_\xi\mu - \mathcal{L}_\eta\lambda + [\lambda,\mu], \mathcal{L}_\xi v - \mathcal{L}_\eta u\big)\,. \tag{79}$$

The minus sign is a convention for these field-space generators. It fixes the plus sign in the ghost differential below.

Let c, χ, υ be odd ghosts and let s be a left odd derivation. All ghosts multiplying component fields are placed on the left. Set

$$sc^\mu = c^\nu\partial_\nu c^\mu, \qquad s\chi = \mathcal{L}_c\chi + \chi^2, \qquad s\upsilon = \mathcal{L}_c\upsilon. \tag{80}$$

On the ordinary fields use the gauge variations with ghosts in place of even parameters, interpreted with this left-placement rule. In particular,

$$s\psi = \mathcal{L}_c\psi + \tfrac{1}{4}\chi^{IJ}\gamma_{IJ}\psi + iq\upsilon\psi, \tag{81}$$

$$s\bar\psi = \mathcal{L}_c\bar\psi - \tfrac{1}{4}\chi^{IJ}\bar\psi\gamma_{IJ} - iq\upsilon\bar\psi. \tag{82}$$

The ordering in the second line matters: both χ and $\bar\psi$ are odd. The expression $-\bar\psi\chi$ is not the same expression. Moving a ghost through an odd dual field changes its sign. The inherited formula with right-placed ghosts is not used without this correction.

Nilpotency follows from the graded Jacobi identity for the parameter algebra and its action on each representation. A simple internal check makes the sign visible. Write the Lorentz representation-valued ghost as $g = \frac{1}{4}\chi^{IJ}\gamma_{IJ}$ and suppress diffeomorphisms and the Abelian factor. Then $sg = g^2$, $s\psi = g\psi$, and the dual rule is $s\bar\psi = \bar\psi g$ when the odd ghost has been moved to the right. Thus

$$s^2\psi = g^2\psi - g(g\psi) = 0, \qquad s^2\bar\psi = (\bar\psi g)g - \bar\psi g^2 = 0. \tag{83}$$

Using $s\bar\psi = -\bar\psi g$ with the same left-derivation convention instead gives $2\bar\psi g^2$. A commuting-parameter covariance test does not detect this Grassmann sign error.

9.3 The classical BV action

For all fields and ghosts Ξ, introduce antifields $\Xi^\ddagger$ with opposite parity and ghost number $-1-\mathrm{gh}(\Xi)$. Use a canonical odd antibracket, with right and left derivatives and the indicated graded ordering. The minimal action can be written

$$S_{\mathrm{BV}} = S_0 + \int \sum_{\Xi} \langle \Xi^\ddagger, s\Xi \rangle. \tag{84}$$

Antisymmetric Lorentz indices are counted once, or equivalently their redundant sum is given its usual factor $1/2$. No higher-antifield terms are needed for this off-shell closed representation algebra.

Proposition 9.1 (Classical master equation)**.** *With the declared graded conventions,* $sS_0 = 0$ *and* $s^2 = 0$ *imply*

$$(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0. \tag{85}$$

Proof. The bracket of S_0 with the antifield-linear terms is twice its gauge variation. The bracket among antifield-linear terms has coefficient $s^2\Xi$ at each antifield, including the ghost antifields. These coefficients vanish. There is no bracket contribution from S_0 with itself. This exhausts the antifield expansion. □

This is a classical construction. It contains no regulated BV Laplacian and establishes no quantum master equation. The odd-field ordering stated above is part of the construction, rather than a claim that the inherited numerical script already checked every Grassmann coefficient.

9.4 Resolved support and Lorentzian reality

Four-dimensional trigonometric polynomials obey the same support lemma as (48). A periodic four-dimensional grid is an algebraic regularization for these identities. It is not a claim that physical spacetime has periodic time or that it is globally hyperbolic. On a slab, the associated boundary terms must instead be retained.

For coefficients in $\mathcal{T}_K^{(4)}$ and a finite Grassmann algebra, the parent action has degree at most six, and the displayed antifield couplings have degree at most three. The master-equation and invariance coefficients have at most seven band-K factors. The sufficient rule $16K < N$ places them below Nyquist and transfers the classical identities coefficientwise. The rule concerns the polynomial parent. It does not automatically survive elimination of inverse tetrads or density square roots.

Reality is built into e and ω. For real Lorentz bivectors, the internal dual obeys $\star^2 = -\mathrm{id}$ and the projectors

$$P_\pm = \tfrac{1}{2}(\mathrm{id} \mp i\star) \tag{86}$$

satisfy $\overline{P_+ X} = P_- X$. Both chiral blocks are retained and related by conjugation. A real vector field with an existing unique flow preserves this real locus. This parent-level statement supplies a real classical starting point. It does not prove a missing finite reality-preserving step for the earlier isolated self-dual coordinates.

10 Boundary constraints and the additional moment-map gate

10.1 An exact time–space split

On $M = I \times \Sigma$ with orientation $\mathrm{d}t \wedge \mathrm{or}(\Sigma)$, write

$$e^I = e^I_\perp \mathrm{d}t + E^I, \qquad \omega^{IJ} = \omega^{IJ}_\perp \mathrm{d}t + \Omega^{IJ}, \qquad a = a_\perp \mathrm{d}t + A. \tag{87}$$

Let $\mathcal{R} = \mathrm{d}_\Sigma \Omega + \Omega \wedge \Omega$ and $\mathcal{F} = \mathrm{d}_\Sigma A$. The curvature and field strength split as

$$R = \mathrm{d}t \wedge (\dot{\Omega} - D_\Omega \omega_\perp) + \mathcal{R}, \qquad F = \mathrm{d}t \wedge (\dot{A} - \mathrm{d}_\Sigma a_\perp) + \mathcal{F}. \tag{88}$$

Define

$$B_{IJ} = \tfrac{1}{4\kappa} \epsilon_{KLIJ} E^K \wedge E^L, \tag{89}$$

$$\Pi = \tfrac{1}{2} \varphi_{IJ} E^I \wedge E^J, \qquad \sigma_I = \tfrac{1}{3!} \epsilon_{IJKL} E^J \wedge E^K \wedge E^L. \tag{90}$$

Direct exterior multiplication gives the spatial symplectic potential

$$\begin{aligned} \Theta_\Sigma = \int_\Sigma [& B_{IJ} \wedge \delta \Omega^{IJ} + \Pi \wedge \delta A \\ & - \tfrac{i}{2} \sigma_I (\bar{\psi} \gamma^I \delta \psi - \delta \bar{\psi} \gamma^I \psi)]. \end{aligned} \tag{91}$$

The minus sign in the spinor term is fixed by the time–space orientation and by moving $\mathrm{d}t$ past the spatial three-form. It must be retained when comparing the spacetime action and its canonical chart.

The temporal coframe multiplies the Einstein–matter three-forms $\mathcal{C}_I = \mathcal{C}^{\mathrm{G}}_I + \mathcal{C}^{\mathrm{M}}_I + \mathcal{C}^{\mathrm{D}}_I$, where

$$\mathcal{C}^{\mathrm{G}}_I = \tfrac{1}{2\kappa} \epsilon_{IJKL} E^J \wedge \mathcal{R}^{KL} - \tfrac{\Lambda}{\kappa} \sigma_I, \tag{92}$$

$$\mathcal{C}^{\mathrm{M}}_I = \varphi_{IJ} \mathcal{F} \wedge E^J - \tfrac{1}{4} \varphi_{KL} \varphi^{KL} \sigma_I, \tag{93}$$

$$\mathcal{C}^{\mathrm{D}}_I = \tfrac{i}{4} \epsilon_{JIKL} E^K \wedge E^L \wedge (\bar{\psi} \gamma^J \mathcal{D}_\Sigma \psi - (\mathcal{D}_\Sigma \bar{\psi}) \gamma^J \psi) - m \sigma_I \bar{\psi} \psi. \tag{94}$$

Spatial integration by parts identifies the Lorentz and electric Gauss forms

$$\mathcal{G}_{IJ} = D_\Omega B_{IJ} - \tfrac{i}{8} \sigma_K \bar{\psi} \{\gamma^K, \gamma_{IJ}\} \psi, \tag{95}$$

$$\mathcal{G}_{\mathrm{U}} = \mathrm{d}_\Sigma \Pi - q \sigma_I \bar{\psi} \gamma^I \psi. \tag{96}$$

Consequently the multiplier equations are $\mathcal{C}_I = \mathcal{G}_{IJ} = \mathcal{G}_{\mathrm{U}} = 0$. These are exact formulas for the split action. The auxiliary constitutive equations and the simplicity relations are additional primary relations; they are not extra first-class gauge generators.

10.2 What is explicit in the internal BFV sector

On a regular symplectic reduction where internal gauge transformations have the stated moment maps, their algebra is Lorentz plus Abelian:

$$\{G_{\mathrm{L}}[\lambda], G_{\mathrm{L}}[\mu]\} = G_{\mathrm{L}}[[\lambda, \mu]], \qquad \{G_{\mathrm{L}}, G_{\mathrm{U}}\} = \{G_{\mathrm{U}}, G_{\mathrm{U}}\} = 0. \tag{97}$$

Its canonical BFV charge is explicit:

$$\Omega_{\text{int}} = G_{\text{L}}[\chi] + G_{\text{U}}[\upsilon] - \tfrac{1}{2}\langle \pi_\chi, [\chi,\chi]\rangle. \tag{98}$$

The moment-map relation cancels the terms quadratic in ghosts, and the Lorentz Jacobi identity cancels the cubic terms. Thus $\{\Omega_{\text{int}}, \Omega_{\text{int}}\} = 0$ under that regular moment-map hypothesis. This finite Lie-algebra mechanism is more explicit than a claim about a complete gravitational boundary charge.

10.3 Normal deformations require more boundary data

For a spatial vector field η, the Cartan identities give

$$\mathcal{L}_\eta \Omega = \iota_\eta \mathcal{R} + D_\Omega(\iota_\eta \Omega), \qquad \mathcal{L}_\eta A = \iota_\eta \mathcal{F} + \mathrm{d}_\Sigma(\iota_\eta A). \tag{99}$$

Accordingly, a covariant tangential generator acquires Lorentz and Abelian compensators to generate an ordinary spatial Lie derivative. The source records the expression

$$J_\Sigma[c,\chi,u] = C[\zeta(c)] + G_{\text{L}}[\chi - \iota_c \Omega] + G_{\text{U}}[u + \iota_c A]. \tag{100}$$

For tangential c, $\zeta(c) = \iota_c E$ is a spatial contraction. For a normal deformation, a spatial one-form does not by itself supply that contraction. One must specify the embedding variables and conjugate momenta, their symplectic extension, and the relevant normal field data. One must then quotient the presymplectic kernel and prove that the gauge action and its Hamiltonian descend.

Proposition 10.1 (Conditional boundary BFV construction)**.** *Suppose a regular derived boundary symplectic space, including the required embedding data, has been constructed. Suppose its full normal and tangential gauge action has an equivariant moment map J_Σ for the declared parameter algebra. Add canonical ghosts and their momenta, and let the ghost Hamiltonian generate the Chevalley–Eilenberg differential in the same conventions. Then their sum Ω_Σ obeys*

$$\{\Omega_\Sigma, \Omega_\Sigma\} = 0. \tag{101}$$

Proof. The bracket of the two moment-map terms reproduces the parameter bracket. The cross bracket with the ghost Hamiltonian cancels that term. The bracket of the ghost Hamiltonian with itself vanishes by the graded Jacobi identity. Equivariance excludes a central defect in the moment-map relation. □

This proposition states the exact missing implication. Off-shell closure of the bulk BRST transformations is insufficient to prove its boundary moment-map hypotheses. The action-split computations establish the split and tangential Cartan identities. They do not construct every normal embedding variable or verify a complete boundary symplectic reduction. The boundary result remains conditional on the hypotheses printed in its statement.

10.4 ADM ghosts and spectral headroom

On a nondegenerate spacelike slice, let $q_{ab} = \eta_{IJ} E^I_a E^J_b$ and define the densitized normal

$$\tilde{n}_I = \tfrac{1}{3!}\epsilon_{IJKL}\epsilon^{abc} E^J_a E^K_b E^L_c. \tag{102}$$

It satisfies $\tilde{n}_I E^I_a = 0$ and $-\eta^{IJ}\tilde{n}_I\tilde{n}_J = \det(q_{ab})$. A parameter $\zeta^I = \nu\tilde{n}^I + \eta^a E^I_a$ yields a densitized lapse ν and shift η. On a correctly reduced boundary space the expected normal bracket has structure shift

$$S^a(\nu,\mu) = \det(q_{cd})q^{ab}(\nu\partial_b\mu - \mu\partial_b\nu). \tag{103}$$

The improved spatial generator differs from the covariant vector generator by its internal compensators. This is the usual reason that the normal bracket may contain both spatial and Gauss terms in the first-order chart.

The map from covariant ghosts to lapse, shift, and internal ghosts depends on E, Ω, A. A canonical lift must transform ghost momenta and, generally, physical momenta as well. It may generate higher ghost terms. Merely replacing the ghosts in a quadratic formula does not prove that the new formula is canonical or nilpotent. The complete charge must be carried through that lift, rather than inferred from its ghost-linear coefficients.

The source estimates at most nine band-K factors in its polynomial ADM representation, so at most sixteen in a bracket. Its sufficient rule is $32K < N$. This explains the additional spectral room required by the densitized normal. It does not discharge the boundary construction, the canonical ghost lift, or rational inverses created after reduction. Those prerequisites are independent of a degree count.

11 A regular real canonical chart

11.1 Time gauge and simplicity pullback

Choose an oriented time-gauge patch

$$E^0 = 0, \qquad E^i = e_a{}^i\,\mathrm{d}x^a, \qquad e := \det(e_a{}^i) > 0. \tag{104}$$

The positive determinant fixes the orientation and ensures invertibility. Define

$$\widetilde{E}^a{}_i = \tfrac{1}{2}\epsilon^{abc}\epsilon_{ijk}e_b{}^j e_c{}^k, \qquad \widetilde{E}^a{}_i e_a{}^j = e\delta_i{}^j. \tag{105}$$

Let $K^i = \Omega^{0i}$ and $\Gamma^i = -\frac{1}{2}\epsilon^i{}_{jk}\Omega^{jk}$. The simplicity pullback gives

$$B_{ij} = 0, \qquad B_{0i} = \tfrac{1}{4\kappa}\epsilon_{ijk}E^j \wedge E^k, \qquad \int_\Sigma B_{IJ}\wedge\delta\Omega^{IJ} = \frac{1}{\kappa}\int_\Sigma \widetilde{E}^a{}_i\delta K_a{}^i\,\mathrm{d}^3x. \tag{106}$$

The antisymmetric IJ sum includes both $0i$ and $i0$, accounting for the factor in the last identity. There is no $\delta\Gamma$ term. Those directions lie in the kernel of the pulled-back gravitational presymplectic form and are fixed by the algebraic connection equation on the regular sector.

Similarly, with $\mathcal{E}^a = \frac{1}{2}\epsilon^{abc}\Pi_{bc}$,

$$\int_\Sigma \Pi\wedge\delta A = \int_\Sigma \mathcal{E}^a\delta A_a\,\mathrm{d}^3x. \tag{107}$$

The Maxwell auxiliary equation then gives its metric Hamiltonian

$$\mathcal{H}_{\mathrm{M}} = \frac{1}{2e}q_{ab}(\mathcal{E}^a\mathcal{E}^b + \mathcal{B}^a\mathcal{B}^b), \qquad \mathcal{B}^a = \tfrac{1}{2}\epsilon^{abc}\mathcal{F}_{bc}. \tag{108}$$

The inverse $1/e$ is well defined in the patch and singular at its boundary.

11.2 Fermionic primary constraints and half densities

Impose the inherited algebraic spinor relation $\bar{\psi} = \psi^\dagger\gamma^0$, where $(\gamma^0)^2 = -\mathrm{id}$. Its phase convention must be included in the physical translation just described. In time gauge $\sigma_0 = e\,\mathrm{d}^3x$ and $\sigma_i = 0$. Equation (91) becomes

$$\Theta_{\mathrm{D}} = \frac{i}{2}\int_\Sigma e(\psi^\dagger\delta\psi - \delta\psi^\dagger\psi)\,\mathrm{d}^3x. \tag{109}$$

In the inherited right–left graded canonical convention, the primary matrix is

$$\{\chi_A(x), \bar{\chi}_B(y)\}_{\mathrm{P}} = -ie(x)\delta_{AB}\delta^3(x-y). \tag{110}$$

It is invertible exactly where $e \neq 0$. Set $\Psi = e^{1/2}\psi$ and $\Psi^\dagger = e^{1/2}\psi^\dagger$. Differentiation gives

$$\Psi^\dagger\delta\Psi = e\psi^\dagger\delta\psi + \tfrac{1}{2}\psi^\dagger\psi\,\delta e, \tag{111}$$

$$\delta\Psi^\dagger\Psi = e\delta\psi^\dagger\psi + \tfrac{1}{2}\psi^\dagger\psi\,\delta e. \tag{112}$$

The determinant terms cancel, so

$$\Theta_{\mathrm{D}} = \frac{i}{2}\int_\Sigma(\Psi^\dagger\delta\Psi - \delta\Psi^\dagger\Psi)\,\mathrm{d}^3x, \qquad \{\Psi_A(x), \Psi^\dagger_B(y)\}_{\mathrm{D}} = -i\delta_{AB}\delta^3(x-y). \tag{113}$$

The half density removes the field-dependent coefficient in the fermionic primary bracket without shifting the gravitational momentum. It requires the chosen real positive square root of e, so it is a local coordinate choice on (104).

11.3 Algebraic torsion and the reduced phase space

For the symmetric Dirac action, define $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $J^I_5 = \bar{\psi}\gamma^I\gamma_5\psi$. In the parent convention, the connection equation has the algebraic solution

$$T^I = \tfrac{\kappa}{4}\epsilon^I{}_{JKL}J^J_5 e^K \wedge e^L, \qquad C^{IJ} = \tfrac{\kappa}{4}\epsilon^{IJ}{}_{KL}J^K_5 e^L, \tag{114}$$

with $\omega = \mathring{\omega}(e) + C$. The solution is unique on the nondegenerate coframe sector. It fixes

$$\Gamma_a{}^i = \mathring{\Gamma}_a{}^i(E) - \tfrac{\kappa}{4}J^0_5 e_a{}^i. \tag{115}$$

One may retain torsion in K and in the substituted connection, or use a torsion-free representation with the induced local four-spinor interaction. In the declared action convention the latter action contribution is $-\frac{3\kappa}{16}\int J_{5I}J^I_5\,\mathrm{vol}_e$. Its sign belongs to that convention; changing the Clifford, orientation, or Dirac-action conventions requires translating it.

The resulting real canonical superphase space has coordinates

$$(K_a{}^i, \widetilde{E}^a{}_i), \qquad (A_a, \mathcal{E}^a), \qquad (\Psi, \Psi^\dagger), \tag{116}$$

and one-form

$$\Theta_{\mathrm{red}} = \int_\Sigma \mathrm{d}^3x\,\left[\frac{1}{\kappa}\widetilde{E}^a{}_i\delta K_a{}^i + \mathcal{E}^a\delta A_a + \frac{i}{2}(\Psi^\dagger\delta\Psi - \delta\Psi^\dagger\Psi)\right]. \tag{117}$$

Its even canonical dimension is 24 per spatial point: eighteen gravitational and six electromagnetic coordinates. The odd primary block has eight components in the stated complexified graded presen-

tation, with the real spinor relation imposed. These counts are local phase-space counts, not a count of globally gauge-quotiented physical degrees of freedom. Rotation, electric, spatial, and Hamiltonian constraints are still first class.

For example, the residual electric constraint is

$$\mathcal{G}_{\mathrm{U}}^{\mathrm{red}} = \partial_a \mathcal{E}^a + q\Psi^\dagger\Psi, \qquad Q[u] = -\int_\Sigma u\mathcal{G}_{\mathrm{U}}^{\mathrm{red}}\,\mathrm{d}^3x. \tag{118}$$

It generates $\delta_u A_a = \partial_a u$ and $\delta_u \Psi = iqu\Psi$. The half-density Lie derivative in the spatial generator is

$$\mathcal{L}_\eta^{(1/2)}\Psi = \eta^a\partial_a\Psi + \tfrac{1}{2}(\partial_a\eta^a)\Psi. \tag{119}$$

This term is forced by the density weight. Treating Ψ as a scalar would give a different moment map.

11.4 Descent and the boundary of the chart

Let ι denote the local solution map for the second-class relations and let $\{\cdot,\cdot\}_{\mathrm{D}}$ be its regular Dirac bracket. If an already constructed boundary BFV vector field is tangent to that second-class surface, and its charge descends through the regular reduction, then

$$\Omega_{\mathrm{red}} = \iota^*\Omega_\Sigma, \qquad \{\Omega_{\mathrm{red}}, \Omega_{\mathrm{red}}\}_{\mathrm{D}} = 0. \tag{120}$$

The implication follows from compatibility of the reduced bracket with the Dirac bracket and the vanishing fine master equation. It does not construct the missing full boundary charge. Fixing time gauge also removes boost gauge directions, so the tangent action must be the residual action or a correctly compensated one.

Degenerate triads, changes of orientation, and singular gauge patches are outside this chart. At $e = 0$ the primary fermionic matrix collapses and the coordinate $e^{1/2}\psi$ ceases to be an invertible change of variables. This is a stratified boundary of the reduction, not a small numerical error. A global reduction would need compatible charts and a treatment of these singular strata.

There is also a spectral boundary. Even when ${e_a}^i$ are trigonometric polynomials, e^{-1}, $e^{1/2}$, and the torsion-free connection are generally not finite Fourier polynomials. The unreduced degree rules $16K < N$ and $32K < N$ therefore cannot simply be reapplied to the reduced chart. One may evaluate its finite-dimensional regular functions directly, or retain auxiliary polynomial variables, but either choice needs its own approximation and support analysis.

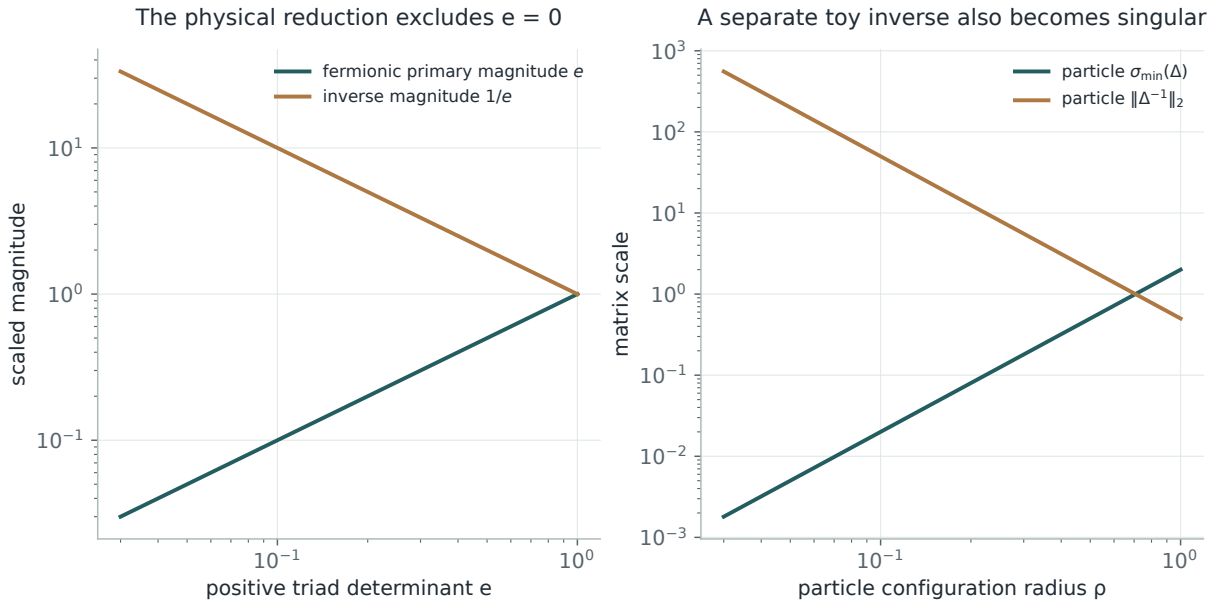


Figure 3. The regular triad sector and a separate finite particle gauge patch. Invertibility is a hypothesis of each reduction, and a singular boundary changes the rank of its constraint matrix.

12 Exactly preserved steps on a finite regular gauge patch

12.1 Endpoint Dirac projection

Consider a finite-dimensional real canonical chart $z \in \mathbb{R}^{2n}$ with constant skew matrix J. Let $\Phi = (\Phi_1, \dots, \Phi_{2m})$ collect independent first-class constraints and regular gauge conditions, or more generally a regular second-class set. Assume these are smooth on a neighborhood of the whole segment between the two step endpoints. For a smooth autonomous function F, define the segment discrete gradient

$$\overline{\nabla} F(z_0, z_1) = \int_0^1 \nabla F((1-s)z_0 + sz_1)\, \mathrm{d}s. \tag{121}$$

The ordinary fundamental theorem of calculus gives

$$F(z_1) - F(z_0) = \overline{\nabla} F(z_0, z_1)^T (z_1 - z_0). \tag{122}$$

Let G have rows $\overline{\nabla}\Phi_j^T$. The endpoint constraint matrix is $\Delta_\Delta = GJG^T$. Require this matrix to be invertible and define

$$J_{\mathrm{D}}^{\Delta} = J - JG^T \Delta_\Delta^{-1} GJ. \tag{123}$$

Direct multiplication proves $GJ_{\mathrm{D}}^{\Delta} = 0$. Since Δ_Δ and its inverse are skew, transposition proves $(J_{\mathrm{D}}^{\Delta})^T = -J_{\mathrm{D}}^{\Delta}$. The endpoint tensor is a projection of the canonical direction into the kernel of all discrete constraint gradients. It need not define a global Poisson bracket or satisfy a Jacobi identity away from the step. Those properties are unnecessary for the preservation argument.

Theorem 12.1 (Exact finite-step preservation)**.** *Let H be a smooth autonomous Hamiltonian. Every exact solution of*

$$z_1 - z_0 = \Delta t\, J_{\mathrm{D}}^{\Delta}(z_0, z_1)\overline{\nabla} H(z_0, z_1) \tag{124}$$

with invertible Δ_{Δ} satisfies

$$\Phi(z_1) = \Phi(z_0), \qquad H(z_1) = H(z_0). \tag{125}$$

Proof. The componentwise chain rule gives $\Phi(z_1) - \Phi(z_0) = G(z_1 - z_0)$, which vanishes because $GJ_{\mathrm{D}}^{\Delta} = 0$. For the Hamiltonian,

$$H(z_1) - H(z_0) = \Delta t\, \overline{\nabla} H^T J_{\mathrm{D}}^{\Delta} \overline{\nabla} H = 0 \tag{126}$$

by skewness. □

The theorem preserves the initial values of the constraints. It does not repair inconsistent initial data. For explicitly time-dependent gauges or $H(z, t)$, ordinary energy is not a conserved quantity. One must formulate the desired balance law or enlarge the phase space by a clock and its conjugate momentum before applying an autonomous preservation theorem.

12.2 A worked particle constraint

In $\mathbb{R}^6$ with $z = (q, p)$ and canonical J, set

$$\Phi_1 = q \cdot q - 1, \qquad \Phi_2 = q \cdot p. \tag{127}$$

At one point the ordinary Dirac matrix is

$$\Delta(z) = \begin{pmatrix} 0 & 2\|q\|^2 \\ -2\|q\|^2 & 0 \end{pmatrix}. \tag{128}$$

It is invertible whenever $q \neq 0$. On the intended constraint surface, q lies on the unit sphere and p is tangent to that sphere. Writing $\bar{q} = (q_0 + q_1)/2$ and $\bar{p} = (p_0 + p_1)/2$, the exact discrete gradients are

$$\overline{\nabla}\Phi_1 = (2\bar{q}, 0), \qquad \overline{\nabla}\Phi_2 = (\bar{p}, \bar{q}). \tag{129}$$

Consequently Δ_{Δ} has the same displayed form with q replaced by $\bar{q}$. Endpoints on the unit sphere can have $\bar{q} = 0$ if they are antipodal. Endpoint admissibility therefore does not imply endpoint-matrix invertibility for an arbitrary large step. Small local steps avoid this singular pair under the regularity hypotheses below.

For $H(q, p) = \frac{1}{2}\|p\|^2 + V(q)$, the segment formula gives an exact discrete gradient of the potential. Polynomial V makes the segment integral a finite algebraic expression. The resulting implicit equations preserve both sphere constraints and H when solved exactly. This is a genuine finite constrained mechanical model. It is a diagnostic for the step mechanism, not an Einstein–Maxwell–Dirac simulation.

12.3 Solver residuals have a precise effect

For an approximate endpoint, define its equation residual by

$$r = (z_1 - z_0) - \Delta t\, J_{\mathrm{D}}^{\Delta}\overline{\nabla} H. \tag{130}$$

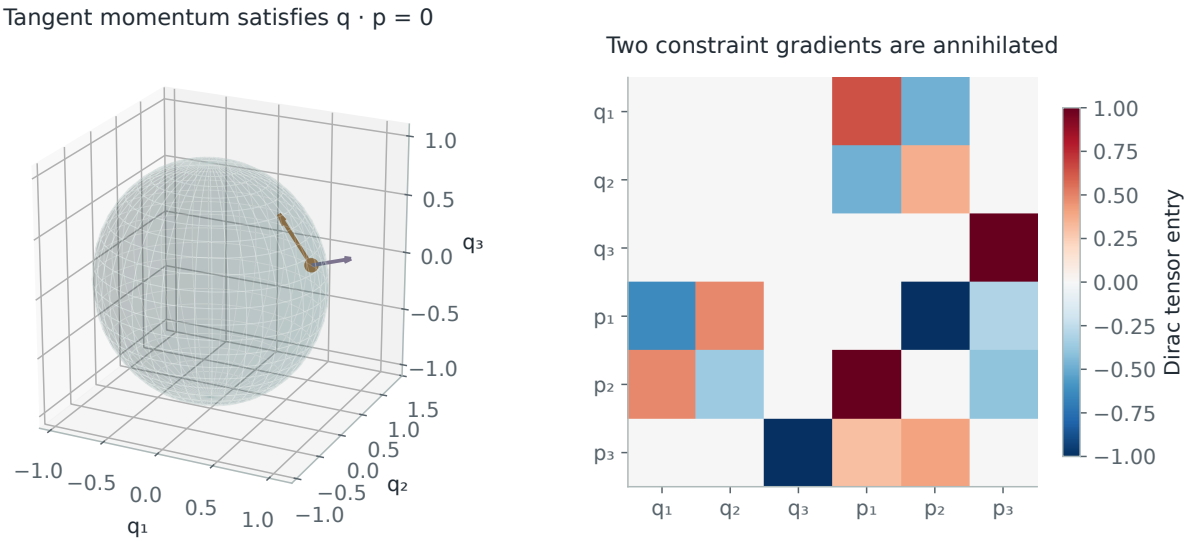


Figure 4. A three-dimensional sphere and its tangent directions, together with the finite Dirac projection. The projection annihilates the specified constraint gradients while retaining physical tangent directions.

If the discrete gradients and tensor are evaluated exactly, the same proof gives

$$\Phi(z_1) - \Phi(z_0) = Gr, \qquad H(z_1) - H(z_0) = \overline{\nabla} H^T r. \tag{131}$$

Thus $\|\Phi(z_1)-\Phi(z_0)\| \le \|G\|\,\|r\|$ and $|H(z_1)-H(z_0)| \le \|\overline{\nabla}H\|\,\|r\|$. These estimates explain why an incompletely solved step need not preserve the invariants. Numerical quadrature of nonpolynomial segment integrals and floating-point linear algebra contribute additional errors and must also be measured. A printed solver tolerance is not the same quantity as the actual equation residual.

The particle certificate records 1000 steps at $\Delta t = 0.1$, with energy error about 7.8×10^{-16}, constraint error about 3.9×10^{-15}, and equation residual about 1.1×10^{-16}. A Runge–Kutta negative control on the continuous Dirac vector field has errors around 10^{-6}. Those values concern that fixed test. They do not establish exact floating-point preservation or a uniform gravity stability theorem. The accompanying atlas reruns a declared finite particle example and also tests incomplete implicit solves.

12.4 Local existence, boundedness, and refinement

At $\Delta t = 0$ and $z_1 = z_0$, the endpoint tensor reduces to the ordinary Dirac tensor if the point constraint matrix is invertible. The derivative with respect to z_1 of the left side of (124) minus its right side is then the identity. If the segment data are smooth in a neighborhood and the inverse matrix stays bounded, the implicit-function theorem gives a unique nearby step for sufficiently small Δt. It does not prove existence of a unique endpoint at arbitrary step size or after leaving that neighborhood.

If the gauge-fixed autonomous Hamiltonian satisfies

$$H(z) \ge a\|z\|^p - b, \qquad a > 0, \quad p > 0, \tag{132}$$

exact preservation yields

$$\|z_n\| \le \left(\frac{H(z_0) + b}{a} \right)^{1/p} \tag{133}$$

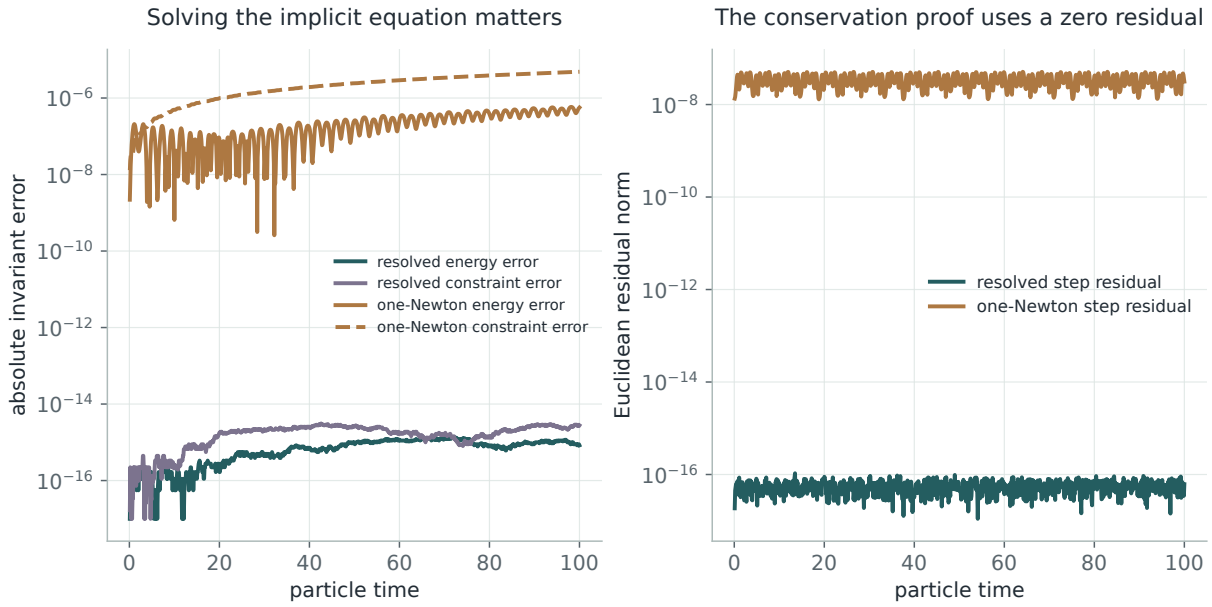


Figure 5. A fresh finite particle experiment contrasts resolved implicit steps with an incomplete-solve control. Its numerical residuals test the preservation mechanism within the specified model and range.

for every continued step in the regular patch. This coercivity is a separate assumption. It is not automatic for a gravitational constraint or a Hamiltonian that vanishes on its constraint surface. Bounded coordinates do not ensure that the gauge matrix stays nonsingular or that the triad determinant stays away from zero.

For an embedding $I_{K\to L}$ and a level-L step U_L, the adaptive map is $U_L \circ I_{K\to L}$. The level-L step preserves the invariants of its embedded initial state. To preserve the intended old data, the embedding must also satisfy the appropriate pullback relations for the Hamiltonian, constraints, gauge conditions, and symplectic form. A generic zero padding or a generic change of gauge does not prove all those relations.

A uniform refinement theorem additionally requires level-independent derivative bounds, inverse Dirac-matrix bounds, coercivity, and a consistency estimate for the actual dynamics. The source leaves those estimates open for the full gravity–matter tower. Strict finite-range implementation is another independent problem: a Fourier derivative couples all grid sites, even though its algebraic operations can be exactly resolved.

13 Elimination, measure balance, and the remaining quantum gate

13.1 A nonlinear Gaussian reduction in closed form

The small elimination model in the transfer chapter explains why an effective action and an effective measure must be transported together. Let $m, n > 0$ and

$$S_F(x_1, x_2, y, z) = \tfrac{1}{2}(x_2 - x_1 y)^2 + \tfrac{m}{2}(y - z)^2 + \tfrac{n}{2} z^2. \tag{134}$$

The fine gauge vector $R_F = \partial_{x_1} + y\partial_{x_2}$ leaves S_F invariant and has zero ordinary divergence. The detail Hessian is

$$M_D = \begin{pmatrix} x_1^2 + m & -m \\ -m & m+n \end{pmatrix}, \qquad D = (m+n)x_1^2 + mn > 0. \tag{135}$$

Writing $\mu = mn/(m+n)$ and $d = x_1^2 + \mu$, the stationary detail values are $y_* = x_1x_2/d$ and $z_* = my_*/(m+n)$. Substitution gives

$$S_C = \frac{\mu x_2^2}{2d}, \qquad R_C = \partial_{x_1} + \frac{x_1x_2}{d}\partial_{x_2}. \tag{136}$$

Differentiation proves $R_C S_C = 0$. Gaussian integration, rather than stationary substitution alone, also gives the density $\rho_C \propto D^{-1/2}$, up to a constant. It satisfies

$$\operatorname{div} R_C = \frac{x_1}{d}, \qquad R_C \log \rho_C = -\frac{x_1}{d}. \tag{137}$$

Their sum vanishes. The effective measure therefore restores the divergence balance of the nonlinear induced gauge transformation.

Eliminating z first gives the effective coefficient μ for y^2 and the factor $(m+n)^{-1/2}$. Eliminating y next gives $d^{-1/2}$. Their product is $D^{-1/2}$, exactly as in direct elimination. For the Euclidean weight $e^{-S/\hbar}$, using a flat coarse measure instead records this determinant as $S_{C,\hbar} = S_C + \frac{\hbar}{2}\log D$, up to an additive constant. A nonlinear symmetry that preserves the classical action can still require a nonconstant induced measure.

13.2 The conditional BV pushforward statement

Suppose a regulated fine quantum BV theory already has compatible odd-symplectic and measure data and satisfies

$$\tfrac{1}{2}(S_F, S_F) - i\hbar\Delta_F S_F = 0. \tag{138}$$

Suppose the variables split into retained and detail sectors with $\Delta_F = \Delta_C + \Delta_D$, and choose an admissible detail Lagrangian $\mathcal{L}_D$. Define the coarse weight by

$$e^{iS_C/\hbar} = \mathcal{N}^{-1} \int_{\mathcal{L}_D} e^{iS_F/\hbar}. \tag{139}$$

When this integral is well defined in the declared analytic or perturbative sense and BV integration by parts has no boundary contribution,

$$\Delta_C e^{iS_C/\hbar} = -\mathcal{N}^{-1} \int_{\mathcal{L}_D} \Delta_D e^{iS_F/\hbar} = 0. \tag{140}$$

This is preservation of the quantum master equation under compatible elimination. It assumes the fine quantum master equation. The classical parent in (84) supplies none of the missing regulator, measure, or anomaly-control data required by (138).

Similarly, direct and iterated elimination agree when the integrals and product gauge fixings admit the relevant Fubini or formal pushforward composition. The quadratic special case is Schur elimination,

$$L_{\text{eff}} = L_{CC} - L_{CD}L_{DD}^{-1}L_{DC}, \tag{141}$$

with the required inverses. A semigroup of elimination maps under these hypotheses does not imply a physically local coarse action. Exact detail elimination can create long-range couplings.

Spherical strong fields

14 Mass coordinates and spherical electrovac

A strong gravitational field is a useful test of a constrained construction because coordinate singularities, physical curvature, and numerical constraint drift can occur at quite different places. The spherical models below keep those distinctions visible. The electrovac model supplies an exact curved spacetime whose coordinates move in time. The scalar model supplies dynamical matter and a numerically observed marginally trapped sphere. Their computational maps have different locality properties, and their evidence supports different conclusions.

The electrovac sector sets the Dirac field to zero and imposes spherical symmetry. It retains a real canonical Einstein–Maxwell system, but it has no nonspherical gravitational radiation. Work in units $G = c = 4\pi\epsilon_0 = 1$. A prime denotes a radial derivative and a dot denotes a time derivative. On a radial interval where $R > 0$ and $\Lambda > 0$, write

$$ds^2 = -N^2dt^2 + \Lambda^2(dr + N^r dt)^2 + R^2 d\Omega^2, \qquad \mathcal{A} = A_r dr + \Phi_e dt. \tag{142}$$

Here R is the areal radius, Λ determines proper radial distance, N is the lapse, and N^r is the radial shift. The Maxwell multiplier is denoted Φ_e to distinguish it from the scalar gradient used later. The bulk canonical action is

$$S = \int dt \int dr \, (P_\Lambda \dot{\Lambda} + P_R \dot{R} + Q\dot{A}_r - N\mathcal{H} - N^r \mathcal{D} - \Phi_e \mathcal{G}) + B_\partial. \tag{143}$$

The boundary term specifies the mass and electromagnetic boundary ensemble. Its treatment cannot be omitted when claiming a global Hamiltonian evolution, even though it does not change the following local bulk identity.

The gauge-improved radial momentum and electric constraints are

$$\mathcal{D} = P_R R' - \Lambda P'_\Lambda, \qquad \mathcal{G} = Q', \tag{144}$$

and the Hamiltonian constraint is

$$\mathcal{H} = -\frac{P_R P_\Lambda}{R} + \frac{\Lambda P_\Lambda^2}{2R^2} + \frac{RR''}{\Lambda} - \frac{RR'\Lambda'}{\Lambda^2} + \frac{R'^2}{2\Lambda} - \frac{\Lambda}{2} + \frac{\Lambda Q^2}{2R^2}. \tag{145}$$

The momenta follow the convention

$$P_\Lambda = -\frac{R}{N}(\dot{R} - N^r R'), \tag{146}$$

$$P_R = -\frac{1}{N}\left[\Lambda(\dot{R} - N^r R') + R(\dot{\Lambda} - (N^r\Lambda)')\right]. \tag{147}$$

These signs fix which Painlevé–Gullstrand branch describes inward future characteristics.

Definition 14.1 (Charged mass coordinate)**.** The charged Misner–Sharp coordinate in this canonical chart is

$$\mathcal{M} = \frac{R}{2}\left[1 - \left(\frac{R'}{\Lambda}\right)^2 + \left(\frac{P_\Lambda}{R}\right)^2\right] + \frac{Q^2}{2R}. \tag{148}$$

The charge contribution is included in $\mathcal{M}$ so that it is constant throughout an electrovac Reissner–Nordström solution.

Proposition 14.2 (Exact triangular mass identity)**.** *For differentiable canonical data in the regular chart,*

$$\mathcal{M}' = -\frac{R'}{\Lambda}\mathcal{H} - \frac{P_\Lambda}{R\Lambda}\mathcal{D} + \frac{Q}{R}\mathcal{G}. \tag{149}$$

Consequently, on a patch where $R'/\Lambda \neq 0$, the systems $(\mathcal{H}, \mathcal{D}, \mathcal{G}) = 0$ and $(\mathcal{M}', \mathcal{D}, \mathcal{G}) = 0$ have the same zero set.

Proof. Differentiating (148) gives

$$\begin{aligned}\mathcal{M}' = \frac{R'}{2} - \frac{R'^3}{2\Lambda^2} - \frac{RR'R''}{\Lambda^2} + \frac{RR'^2\Lambda'}{\Lambda^3} + \frac{P_\Lambda P_\Lambda'}{R} - \frac{P_\Lambda^2 R'}{2R^2} \\ + \frac{QQ'}{R} - \frac{Q^2R'}{2R^2}.\end{aligned} \tag{150}$$

Expanding the right side of (149) cancels the two P_R terms and produces precisely this expression. The constraint transformation is upper triangular with diagonal $(-R'/\Lambda, 1, 1)$, so it is invertible under the stated condition. □

This calculation changes constraint coordinates before discretization. It does not prove closure of an arbitrary pointwise discretization of the original hypersurface-deformation algebra. It also establishes equivalence of zero sets, rather than proving that every Poisson bracket of the replacement functions vanishes. The distinction matters when the reduced construction is compared with a general BFV system.

15 Horizon penetration and the regular radial chart

Fix $M > 0$, a real charge Q, and a branch $\varepsilon \in \{+1, -1\}$. On $2MR - Q^2 > 0$, define

$$p(R) = \varepsilon\sqrt{2MR - Q^2}, \qquad v(R) = \frac{p(R)}{R}. \tag{151}$$

The Painlevé–Gullstrand metric takes the form

$$ds^2 = -dt_P^2 + (dR + v(R)dt_P)^2 + R^2 d\Omega^2. \tag{152}$$

No inverse factor of $1 - 2M/R + Q^2/R^2$ appears. The metric can therefore cross a horizon while the lapse and radial spatial metric remain regular.

Take a smooth radial map $R = F(t, r)$ with $F_r > 0$ and identify $t_P = t$. Expanding $dR = F_t dt + F_r dr$ identifies

$$N = 1, \qquad \Lambda = F_r, \qquad N^r = \frac{F_t + v(F)}{F_r}. \tag{153}$$

The corresponding canonical data are

$$P_\Lambda = p(F), \qquad P_R = \frac{MF_r}{p(F)}, \qquad A_r = 0, \quad \Phi_e = -\frac{Q}{F}, \quad Q' = 0. \tag{154}$$

The sign in P_R is contained in $p(F)$.

Proposition 15.1 (A time-dependent exact gauge family)**.** *The data* (154) *satisfy the local electrovac constraints and canonical evolution equations wherever $F_r > 0$, $F > 0$, and $2MF - Q^2 > 0$.*

Proof. First, $R'/\Lambda = 1$ and $p(F)^2 = 2MF - Q^2$ imply $\mathcal{M} = M$. Since

$$P'_\Lambda = \frac{MF_r}{p(F)} = P_R, \tag{155}$$

the momentum constraint vanishes. The mass identity and $Q' = 0$ then give $\mathcal{H} = 0$.

For evolution, the radius equation gives

$$\dot{R} = -\frac{P_\Lambda}{R} + N^r R' = -v(F) + F_t + v(F) = F_t. \tag{156}$$

The radial-metric equation is

$$\dot{\Lambda} = \frac{\Lambda P_\Lambda}{R^2} - \frac{P_R}{R} + (N^r \Lambda)'. \tag{157}$$

Use $p_R = M/p$ and $v_R = M/(pR) - p/R^2$. The non-F_{tr} terms cancel, leaving $\dot{\Lambda} = F_{tr}$. The momentum equations in the $N = 1$ convention are

$$\dot{P}_\Lambda = N^r P'_\Lambda + \frac{1}{2}\left[1 - \frac{P_\Lambda^2 + Q^2}{R^2} - \left(\frac{R'}{\Lambda}\right)^2\right], \tag{158}$$

$$\dot{P}_R = \frac{\Lambda(P_\Lambda^2 + Q^2)}{R^3} - \frac{P_\Lambda P_R}{R^2} + (N^r P_R)' - \frac{R''}{\Lambda} + \frac{\Lambda' R'}{\Lambda^2}. \tag{159}$$

Substituting the data and differentiating $p(F)^2 = 2MF - Q^2$ reduces their right sides to

$$\frac{MF_t}{p(F)}, \qquad M\left[\frac{F_{tr}}{p(F)} - \frac{MF_r F_t}{p(F)^3}\right], \tag{160}$$

which are the direct time derivatives of the two momenta. Finally, $\Phi'_e = QF_r/F^2$ cancels $N\Lambda Q/R^2$ in $\dot{A}_r = N\Lambda Q/R^2 - \Phi'_e$. □

An arbitrary admissible F therefore creates coordinate motion through one fixed electrovac spacetime. Neither the mass nor the charge changes. A numerical reconstruction of this family is a nonlinear strong-field consistency test, but it is not matter collapse.

15.1 Charged horizons and the turning surface

Write

$$f(R) = 1 - \frac{2M}{R} + \frac{Q^2}{R^2}. \tag{161}$$

For $0 < |Q| < M$, its two positive roots are

$$r_{\pm} = M \pm \sqrt{M^2 - Q^2}. \tag{162}$$

At either root $p(R)^2 = R^2$, so the canonical quantities in (154) remain finite if F_r does. For $|Q| = M$, the two roots coincide at $R = M$. For $|Q| > M$ there is no real root. At $Q = 0$, the outer root is $2M$ and the formal inner root 0 is the singular endpoint, not a second regular horizon.

The square-root chart instead meets a turning surface at

$$R_{\text{turn}} = \frac{Q^2}{2M}. \tag{163}$$

There $p = 0$, and $P_R = MF_r/p$ becomes unbounded for nonzero F_r. This is a limitation of the chosen flat spatial slice and canonical chart. Horizon penetration does not remove the need for a positive distance from this turning surface. It also does not remove the physical curvature singularity at $R = 0$.

The invariant curvature diagnostic is

$$\mathcal{K} = \frac{48M^2}{R^6} - \frac{96MQ^2}{R^7} + \frac{56Q^4}{R^8}. \tag{164}$$

For $M = 1$, $Q = 0.8$, the radii are $r_+ = 1.6$, $r_- = 0.4$, and $R_{\text{turn}} = 0.32$. The benchmark interval $R \in [0.6, 6]$ crosses the outer horizon only. At its inner end $\mathcal{K} \simeq 199.6646853$, while $\mathcal{K}(r_+) \simeq 1.106262207$. High curvature and regular horizon-crossing coordinates coexist on this interval.

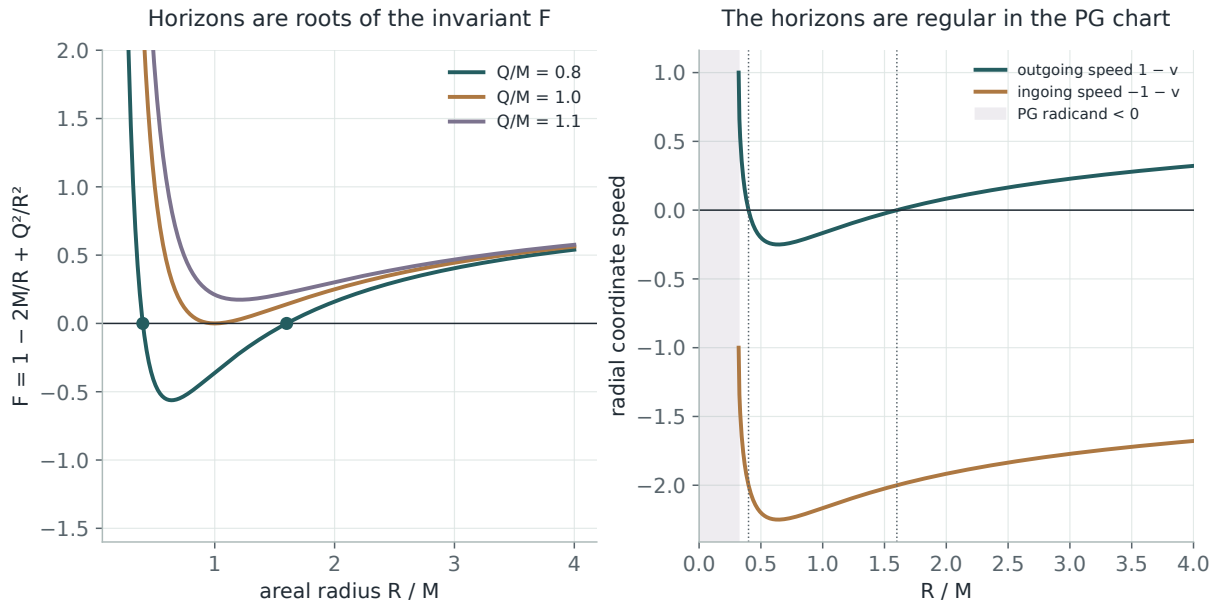


Figure 6. The horizon equation and the domain of the charged Painlevé–Gullstrand chart are different conditions. The benchmark crosses the outer horizon while remaining above the turning surface. The extremal and superextremal curves illustrate the change in the number and multiplicity of horizon roots.

15.2 Horizon conditioning

For a subextremal outer horizon, put $d = \sqrt{M^2 - Q^2} > 0$. Then

$$f'(r_+) = \frac{2d}{r_+^2}. \tag{165}$$

If a nearby continuous numerical profile has a selected root near this simple root, the local inverse estimate gives

$$|\delta r_+| \le \frac{r_+^2}{2d}\, \|\delta f\|_\infty + O(\|\delta f\|_\infty^2). \tag{166}$$

The estimate describes the nearby root and presupposes its localization to a region where the exact derivative is nonzero. It does not make every zero of an arbitrarily oscillatory perturbation physically relevant. The factor is $2.133333333\ldots$ in the benchmark and diverges as $d \downarrow 0$.

At extremality $f(R) = (1 - M/R)^2$ has a double root. A perturbation of size ϵ can move the root by order $\sqrt{\epsilon}$ or remove it entirely. Finite canonical data at a horizon therefore do not imply uniformly well-conditioned horizon detection. The first trapped sphere in the scalar calculation below also occurs at a spatial compactness maximum, and must not be assigned this simple-root estimate without checking its event geometry.

16 Finite radial reconstruction and its stability boundary

On interior points of a uniform radial grid, use the fourth-order derivative

$$(D_4 u)_j = \frac{-u_{j+2} + 8u_{j+1} - 8u_{j-1} + u_{j-2}}{12h}. \tag{167}$$

Its stencil radius is two. Given M, Q, and a fixed branch ε, a radius array satisfying $D_4 R_j > 0$ and $2MR_j - Q^2 > 0$ determines

$$\Pi_h[R]_j = \left(R_j, D_4 R_j, \varepsilon\sqrt{2MR_j - Q^2}, \frac{M D_4 R_j}{\varepsilon\sqrt{2MR_j - Q^2}}, Q \right). \tag{168}$$

This is a local reconstruction of $(R, \Lambda, P_\Lambda, P_R, Q)$, rather than an orthogonal Hilbert-space projector. If it is extended to a full state by discarding the other variables and retaining R, it is idempotent on its admissible image.

Proposition 16.1 (Exact finite reduced constraints)**.** *On the image of* (168)*, the four reduced constraints*

$$\begin{aligned} C_{\Lambda,j} &= \Lambda_j - D_4 R_j, & C_{M,j} &= P_{\Lambda,j}^2 - 2MR_j + Q^2, \\ C_{P,j} &= P_{R,j}P_{\Lambda,j} - M\Lambda_j, & C_{Q,j} &= Q_j - Q \end{aligned} \tag{169}$$

vanish in exact arithmetic. The discrete mass coordinate obtained by replacing R' with $D_4 R$ in (148) *equals M at every reconstructed node.*

Proof. Each constraint vanishes by direct substitution. In the discrete mass coordinate $D_4 R/\Lambda = 1$ and $P_\Lambda^2 = 2MR - Q^2$, leaving $\frac{R}{2}(2M/R - Q^2/R^2) + Q^2/(2R) = M$. □

These identities are exact algebraic constraints on the reconstruction manifold. The original ADM expressions evaluated with finite differences have a different status: the stencil does not obey a pointwise chain rule. Thus $D_4\sqrt{2MR - Q^2}$ need not equal $MD_4R/\sqrt{2MR - Q^2}$ exactly. The original momentum and Hamiltonian constraints are retained as continuum-consistency diagnostics, rather than being relabeled as exact finite invariants.

16.1 A level-independent reconstruction estimate

Assume a uniform regular set

$$R_j \geq R_* > 0, \quad 2MR_j - Q^2 \geq p_*^2 > 0, \quad 0 < d_* \leq D_4 R_j \leq d^*. \tag{170}$$

For two arrays in this set, with the same (M, Q, ε), differentiation along the segment between them gives

$$\|\delta\Lambda\|_\infty \leq \|D_4 \delta R\|_\infty , \tag{171}$$

$$\|\delta P_\Lambda\|_\infty \leq \frac{M}{p_*} \|\delta R\|_\infty , \tag{172}$$

$$\|\delta P_R\|_\infty \leq \frac{M}{p_*} \|D_4 \delta R\|_\infty + \frac{M^2 d^*}{p_*^3} \|\delta R\|_\infty . \tag{173}$$

For example, $\partial_R p = M/p$ and the two partial derivatives of $M\Lambda/p$ have magnitudes at most M/p_* and $M^2 d^*/p_*^3$. The admissible segment preserves the displayed linear radius and derivative bounds. The constants are independent of h.

The input norm contains a radial derivative. A small pointwise radius error alone therefore does not establish a small error in every reconstructed canonical field. This estimate transfers an already controlled discrete $W^{1,\infty}$ radius error to the reconstructed state. It does not create that derivative control from an L^2 estimate.

16.2 Direct and split radius evolution

On the positive branch, restriction of the radius equation to the reconstruction manifold gives

$$\dot{R}_j = g(R_j) + b_j (D_4 R)_j, \qquad g(R) = -\frac{\sqrt{2MR - Q^2}}{R}, \tag{174}$$

where $b = N^r$ is prescribed in the electrovac benchmark. A split alternative is

$$\dot{R} = g(R) + A_h[b]R, \qquad A_h[b]u = \frac{1}{2}(bD_4u + D_4(bu) - (D_4b)u). \tag{175}$$

On smooth data these formulas differ by half of the discrete product-rule defect, which is fourth order away from boundaries. Their exact finite-grid operators differ.

The archived implementation advances the radius explicitly with RK4. The map Π_h can reconstruct the other fields from each admissible stage radius, and its reduced identities hold on that reconstructed image. At output, however, the code overwrites two canonical boundary nodes on each side with exact manufactured data. Those boundary nodes are not generally the exact image of the finite derivative reconstruction. Its reported constraint norms exclude four nodes from each end. This procedure does not use an implicit discrete-gradient slab solve. Conservation by an implicit invariant-preserving method elsewhere in this monograph remains conditional on actually solving its nonlinear step equation. It cannot be used as evidence that this explicit radius algorithm preserves the original ADM Hamiltonian exactly.

Proposition 16.2 (Split-form energy identity)**.** *Let D be skew adjoint for the grid inner product $\langle u, w\rangle_h$, and let multiplication by the real array b be self-adjoint. Then*

$$\langle u, A_h[b]u\rangle_h = -\frac{1}{2}\langle u, (Db)u\rangle_h . \tag{176}$$

For two solutions of the split equation with the same prescribed b, remaining in a positive-radicand interval where $|g'| \le L_g$, one obtains

$$\left\|R(t) - \widetilde{R}(t)\right\|_h \le e^{(L_g + S_1/2)t}\left\|R(0) - \widetilde{R}(0)\right\|_h , \qquad \|Db\|_\infty \le S_1. \tag{177}$$

Proof. Skew adjointness cancels $\langle u, bDu\rangle_h + \langle u, D(bu)\rangle_h$. The remaining term gives (176). With $e = R - \widetilde{R}$, the nonlinear difference satisfies $|g(R_j) - g(\widetilde{R}_j)| \le L_g|e_j|$, where

$$g'(R) = \frac{MR - Q^2}{R^2\sqrt{2MR - Q^2}}. \tag{178}$$

Thus $\frac{1}{2}\partial_t \|e\|_h^2 \le (L_g + S_1/2)\|e\|_h^2$, and integration gives the claim. □

The skew-adjoint hypothesis needs care on a physical radial interval. A periodic D_4 satisfies it, but a globally periodic radius array cannot also satisfy $D_4R_j > 0$ everywhere: summing the derivative over one period gives zero. The periodic identity is an exact algebraic check of the split stencil. It does not by itself give a nonvacuous stability theorem for the globally monotone spherical reconstruction.

For an interval summation-by-parts derivative with a diagonal positive norm matrix H and $HD + D^TH = B$, multiplication by b commutes with H. The uncancelled term is instead

$$\langle u, A_h[b]u\rangle_H = \frac{1}{2}u^TB\ \mathrm{diag}(b)u - \frac{1}{2}\langle u, (Db)u\rangle_H . \tag{179}$$

Its boundary flux must be controlled using an actual boundary closure and appropriate boundary data. Saying that a stencil is centered in the interior does not establish this property for the complete interval operator.

The periodic Fourier symbol is

$$\widehat{D}_4(\theta) = \frac{i}{h}\frac{8\sin\theta - \sin 2\theta}{6}. \tag{180}$$

Its maximum dimensionless magnitude is $1.3722219798\ldots$. Applying the RK4 imaginary-axis stability interval to frozen constant advection gives

$$\frac{|b|\Delta t}{h} \le 2.0612023174\ldots. \tag{181}$$

This is a frozen-coefficient diagnostic. A full variable-coefficient nonlinear RK4 convergence theorem still requires a uniform stability argument for the implemented boundary closure and a suitable stronger norm. Fourth-order consistency, the semidiscrete L^2 estimate, and this one CFL number do not alone prove fourth-order $W^{1,\infty}$ convergence.

17 What the electrovac benchmark measures

The retained run uses

$$F(t,r) = r + 0.08\sin(0.7t)\sin^2\left(\pi\frac{r-0.6}{5.4}\right), \quad 0 \le t \le 0.4, \tag{182}$$

with $M = 1$, $Q = 0.8$, and $\Delta t/h$ close to 0.1. The sampled minimum radial Jacobian is 0.9871380683 and the minimum momentum radicand is 0.56. The exact shift, ghost values, and two boundary nodes on either side are supplied from the manufactured family. Errors are measured on interior nodes after excluding four points from each end.

The recorded state errors are

N	$\|\delta R\|_\infty$	$\|\delta\Lambda\|_\infty$	$\|\delta P_\Lambda\|_\infty$	$\|\delta P_R\|_\infty$
64	$8.3247\,10^{-9}$	$4.1362\,10^{-8}$	$5.7367\,10^{-9}$	$2.3732\,10^{-8}$
128	$5.0495\,10^{-10}$	$2.5079\,10^{-9}$	$3.3079\,10^{-10}$	$1.3795\,10^{-9}$
256	$3.1077\,10^{-11}$	$1.5438\,10^{-10}$	$2.0248\,10^{-11}$	$8.4854\,10^{-11}$
512	$1.9296\,10^{-12}$	$1.0485\,10^{-11}$	$1.2559\,10^{-12}$	$5.3844\,10^{-12}$

The least-squares orders are respectively 3.99856, 3.96010, 4.02393, and 4.00788. The ordinary ADM Hamiltonian on exact projected data has fitted order 3.70410. These are results for a smooth, forced trajectory on four grids. The tiny final errors are close enough to roundoff that they should not be interpreted as arbitrarily extendible asymptotic scaling laws.

On the tested interior of the $N = 512$ reconstructed grid, the retained mass residual is $4.44\,10^{-16}$, the flat-slice residual is zero, the algebraic mass constraint is $2.22\,10^{-15}$, and the reduced momentum residual is $1.11\,10^{-16}$. These floating-point numbers are consistent with the exact algebraic reconstruction theorem. They are not an exact-zero assertion about floating arithmetic or the overwritten boundary nodes.

The negative control evolves all four canonical fields without the reconstruction. At $N = 128$, it reports mass drift $2.25125\,10^{-4}$, radial momentum residual $1.27688\,10^{-3}$, and Hamiltonian residual $1.25735\,10^{-2}$. This gives the reduced reconstruction a meaningful comparator. It does not show that every other constrained algorithm fails or that the direct ADM system is ill posed.

A smooth perturbation of amplitude 10^{-8} evolved with the split form has recorded L^2 amplification between 1.03345462 and 1.03348106 on $N = 128, 256, 512, 1024$. The spread is $2.64438\,10^{-5}$. This supports the bounded experiment's level-insensitive behavior. It leaves the interval boundary and stronger-norm theorem obligations visible.

The outer horizon remains at the fixed areal radius 1.6. Its computational coordinate moves because F changes. The retained final coordinate comparison at $N = 256$ gives $|r_{+,h} - r_+^{\text{exact}}| = 1.89262\,10^{-11}$. This measures tracking of a known horizon under coordinate deformation. It does not measure the formation or growth of a black hole.

The electrovac verification record contains nineteen passed checks. The symbolic identity and archived data structure have supporting audit records, but these are distinct from a replay of the electrovac evolution. The numerical table is retained as recorded evidence. Its environment and command provenance are specified in the verification appendix.

18 A dynamical spherical scalar system

To test dynamical concentration of matter, replace electrovac by a real massless scalar. Use $G = c = 1$ and the rescaled field $\varphi = \sqrt{4\pi}\,\phi_{\text{phys}}$. The scalar canonical pair is (φ, P_φ), and the matter additions to the Hamiltonian and momentum constraints are

$$\mathcal{H}_{\text{matt}} = \frac{P_\varphi^2}{2\Lambda R^2} + \frac{R^2\varphi'^2}{2\Lambda}, \qquad \mathcal{D}_{\text{matt}} = P_\varphi\varphi'. \tag{183}$$

The gravitational part is (145) with $Q = 0$.

Impose $R = r$ and $\Lambda = 1$, and set

$$N = \sigma > 0, \quad N^r = \sigma v, \quad P_\Lambda = rv, \quad P_\varphi = r^2\Pi, \quad \Phi = \varphi'. \tag{184}$$

Here Π is the scalar momentum per areal area, not the reconstruction map Π_h. Define

$$m = \frac{r}{2}v^2, \qquad v = \sqrt{\frac{2m}{r}}. \tag{185}$$

The inward branch gives

$$ds^2 = -\sigma^2 dt^2 + (dr + \sigma v dt)^2 + r^2 d\Omega^2. \tag{186}$$

The following derivation applies on smooth patches with $r > 0$, $\sigma > 0$, and $v > 0$. These hypotheses keep every division legitimate.

The momentum constraint removes P_R algebraically:

$$P_R = P_\Lambda' - P_\varphi\Phi. \tag{187}$$

Since $m' = P_\Lambda P_\Lambda'/r - P_\Lambda^2/(2r^2)$, inserting this relation into the Hamiltonian constraint gives

$$m' = r^2\mathcal{E}, \qquad \mathcal{E} = \frac{1}{2}(\Pi^2 + \Phi^2) + v\Pi\Phi. \tag{188}$$

Preservation of $\Lambda = 1$ gives

$$0 = \sigma\left(\frac{v}{r} - \frac{P_R}{r}\right) + (\sigma v)' = \sigma r\Pi\Phi + \sigma' v, \tag{189}$$

hence

$$(\log\sigma)' = -\frac{r\Pi\Phi}{v}. \tag{190}$$

The scalar evolution equations are

$$\dot{\varphi} = \sigma(\Pi + v\Phi), \tag{191}$$

$$\dot{\Phi} = \partial_r[\sigma(\Pi + v\Phi)], \tag{192}$$

$$\dot{\Pi} = r^{-2}\partial_r[r^2\sigma(\Phi + v\Pi)]. \tag{193}$$

If $\Phi = \varphi'$ initially and compatible boundary data are imposed, the first two equations preserve that relation in the continuum.

At $v = 0$, the lapse equation requires a separately specified extension. Smoothness of Π and Φ alone does not establish that $r\Pi\Phi/v$ is removable. A vacuum interval with vanishing scalar data can provide an appropriate extension, but that must be part of the argument. The Gaussian benchmark has very small tails rather than exact compact support, and its numerical treatment of small v is described below. This calculation does not prove a regular-center theorem.

18.1 Exact mass flux and compatibility

Put $A = \Pi + v\Phi$ and $B = \Phi + v\Pi$. The remaining Einstein momentum evolution gives

$$\dot{m} = r^2\mathcal{J}, \qquad \mathcal{J} = \sigma AB. \tag{194}$$

One can verify its origin directly from the P_Λ equation. In the flat-slice gauge, the gravitational terms give $\sigma v P_\Lambda' - \sigma v^2/2 - r\sigma'$. The lapse-gradient term is essential. The matter variation contributes $\sigma r^2(\Pi^2 + \Phi^2)/2$. As $\dot{m} = v\dot{P}_\Lambda$, their sum becomes

$$\dot{m} = \sigma v\left[m' + \frac{r^2}{2}(\Pi^2 + \Phi^2)\right] - rv\sigma'. \tag{195}$$

Using (188) and (190) yields $\dot{m} = r^2\sigma[v(\Pi^2 + \Phi^2) + (1 + v^2)\Pi\Phi]$, which is (194).

Proposition 18.1 (Continuum mass-constraint propagation)**.** *On the regular patch, the scalar evolution, lapse relation, and mass-flux evolution satisfy*

$$\partial_t(r^2\mathcal{E}) = \partial_r(r^2\mathcal{J}). \tag{196}$$

Thus $\partial_t(m' - r^2\mathcal{E}) = 0$. A radial mass constraint imposed initially propagates under this continuum system.

Proof. Expanding (193) gives $\dot{\Phi} = (\sigma A)'$ and $\dot{\Pi} = (\sigma B)' + 2\sigma B/r$. Differentiate $\mathcal{E}$ and subtract the radial derivative of the flux. Terms involving Π', Φ', and v' cancel, leaving

$$\partial_t(r^2\mathcal{E}) - \partial_r(r^2\mathcal{J}) = r^2[\sigma' AB + \Pi\Phi\dot{v}]. \tag{197}$$

Since $m = rv^2/2$, the flux equation implies $\dot{v} = r\sigma AB/v$. The remaining expression is

$$r^2 AB\left[\sigma' + \frac{\sigma r\Pi\Phi}{v}\right] = 0 \tag{198}$$

by (190). This establishes the identity without assuming the radial constraint in its last cancellation step. □

On a fixed radial interval $[r_i, r_o]$, the exact balance is

$$\frac{d}{dt}(m(t, r_o) - m(t, r_i)) = r_o^2\mathcal{J}(t, r_o) - r_i^2\mathcal{J}(t, r_i). \tag{199}$$

Boundary fluxes, rather than a statement that a finite-domain mass must be exactly constant, are the appropriate continuum reference. The numerical prefix solve below does not satisfy this identity algebraically at every finite grid and time step.

18.2 Characteristic channels and the loss of a coordinate energy bound

Define $w_+ = \Pi + \Phi$ and $w_- = \Pi - \Phi$. Their equations are

$$\dot{w}_+ = \sigma(v+1)w_+' + [\sigma v' + \sigma'(v+1)]w_+ + \frac{\sigma}{r}[(v+1)w_+ + (v-1)w_-], \tag{200}$$

$$\dot{w}_- = \sigma(v-1)w_-' + [\sigma v' + \sigma'(v-1)]w_- + \frac{\sigma}{r}[(v-1)w_- + (v+1)w_+]. \tag{201}$$

Written as transport equations $\dot{w} + c\,w' =$ source, their coordinate speeds are

$$c_+ = -\sigma(v+1), \qquad c_- = \sigma(1-v). \tag{202}$$

The plus channel is inward on the positive branch. The minus channel is outward when $v < 1$, stationary at $v = 1$, and inward when $v > 1$.

The same channels give the exact algebraic identities

$$\mathcal{E} = \frac{1}{4}[(1+v)w_+^2 + (1-v)w_-^2], \tag{203}$$

$$\mathcal{J} = \frac{\sigma}{4}[(1+v)^2 w_+^2 - (1-v)^2 w_-^2]. \tag{204}$$

For $0 \le v < 1$, $\mathcal{E} \ge (1-v)(w_+^2 + w_-^2)/4$. This lower bound degenerates at a horizon. For $v > 1$, the outward-channel coefficient in this coordinate mass density is negative. The scalar's local normal-frame quadratic density $(\Pi^2 + \Phi^2)/2$ remains nonnegative. The loss of coercivity in $\mathcal{E}$ therefore diagnoses this slicing's mass-coordinate energy estimate, rather than negative physical scalar energy.

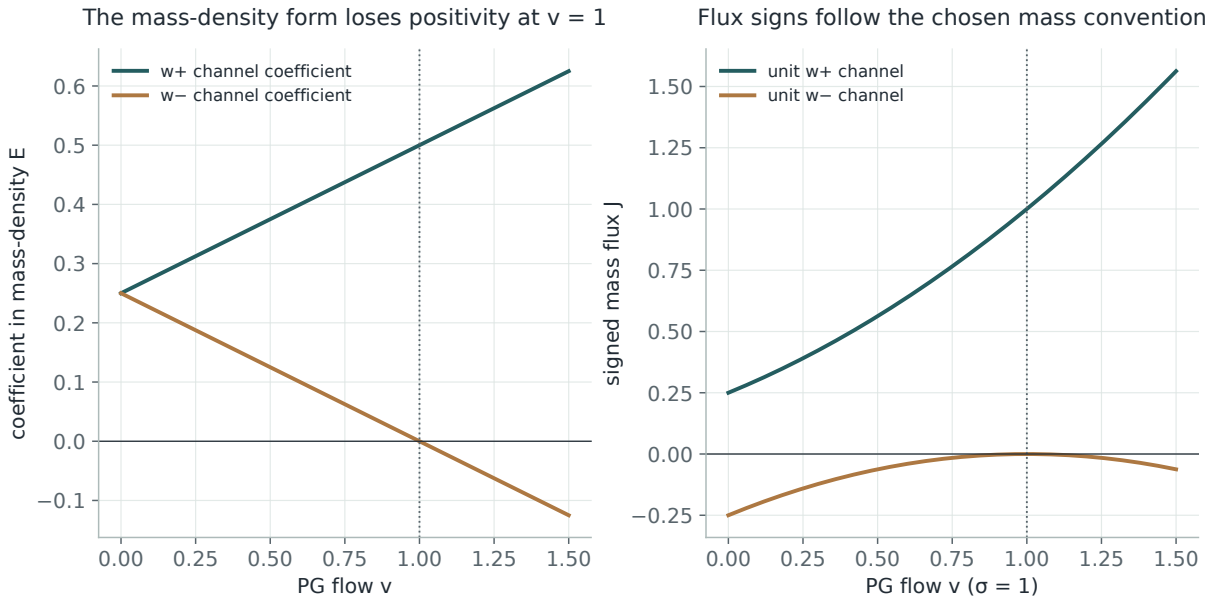


Figure 7. The mass density and mass flux resolve into the two scalar characteristic channels. The outgoing characteristic and its contribution to the coordinate mass density degenerate at $v = 1$. These are exact algebraic profiles, not a newly evolved collapse solution.

19 Trapped spheres and the finite-stencil scalar experiment

For (186), choose future radial null vectors

$$\ell_{\pm} = \sigma^{-1}\partial_t + (-v \pm 1)\partial_r. \tag{205}$$

Because the sphere area is $4\pi r^2$, its expansion along ℓ is $2\ell(r)/r$. Consequently

$$\theta_{+} = \frac{2}{r}(1 - v), \qquad \theta_{-} = -\frac{2}{r}(1 + v). \tag{206}$$

On the positive branch, $v = 1$ is a marginally outer trapped sphere, and $v > 1$ gives both expansions negative. Equivalently, the compactness $2m/r$ equals or exceeds one. The normalization of the null vectors affects expansion magnitude but not these signs. A marginally trapped sphere is a local geometric diagnostic. It is not by itself identification of a global event horizon.

19.1 What is local in the algorithm

At each RK4 stage the matter channels determine the geometry through two radial solves. The mass constraint is integrated outward with $m(r_{\text{exc}}) = 0$. The lapse derivative is integrated and the result normalized by $\sigma(r_{\max}) = 1$. These are triangular prefix operations. A matter value near the inner end can affect the reconstructed mass and lapse far along the interval in the same stage. The full stage map is therefore globally coupled.

The characteristic evolution uses the second-order one-sided derivatives

$$(D_{+}u)_j = \frac{-3u_j + 4u_{j+1} - u_{j+2}}{2h}, \qquad (D_{-}u)_j = \frac{3u_j - 4u_{j-1} + u_{j-2}}{2h}. \tag{207}$$

The inward channel uses D_+ and the outgoing channel uses D_- outside a horizon, switching to D_+ inside. These interior matter stencils have radius two. The source also uses a radius-two fourth-order interior derivative for v' and separate one-sided boundary formulas. Finite stencil radius for these derivatives must not be promoted to a finite dependency cone for the geometry-coupled step.

The mass integration uses radial RK4 with interpolated midpoint fields, followed by Simpson-type lapse quadrature. These are approximate quadratures. They are neither exact mass-constraint algebraic reconstruction nor an implicit invariant-preserving solver. The scalar spatial evolution is second order despite its fourth-order time integrator.

Several implementation conventions are mathematically material. The mass right side and the reconstructed v use $\sqrt{\max(2m/r, 0)}$. The lapse derivative is set to zero where $v \le 10^{-13}$. Incoming characteristic values are explicitly set after a completed RK4 step, with separate boundary formulas during stages. These choices guard a finite calculation near vacuum and boundaries. An exact continuum proof on $v > 0$ does not establish their uniform consistency as the threshold is varied or $h \downarrow 0$.

19.2 Initial data and boundary scope

The numerical domain is $[r_{\text{exc}}, r_{\text{max}}] = [0.05, 30]$. It replaces a regular center by a small excised core. The characteristic data are $w_-(r_{\text{exc}}) = 0$ and $w_+(r_{\text{max}}) = 0$. These prohibit prescribed incoming radiation from the inner and outer boundaries in the respective channels. The radial mass boundary condition additionally fixes the excised mass to zero at every geometry solve.

The initial scalar packet is

$$\varphi(0, r) = Ae^{-(r-10)^2}, \qquad \Phi(0, r) = \partial_r \varphi(0, r), \qquad \Pi(0, r) = \Phi(0, r) + \frac{\varphi(0, r)}{r}. \tag{208}$$

The last equation is the flat-space ingoing condition for $\chi = r\varphi$, used to initialize the scalar. The nonlinear gravitational constraints are then solved. It is therefore an ingoing packet prescription, rather than exactly freezing the geometry to Minkowski space.

The strong amplitude is $A = 0.10$ and the control amplitude is $A = 0.05$. The strong event occurs near $r = 3.1$, well separated from both boundaries in coordinate distance. That separation is useful but is not a substitute for studying boundary influence: the geometry solve is global, and finite-domain mass drift can contain both boundary and discretization effects.

19.3 Recorded first marginally trapped sphere

The event finder takes a quadratic interpolation of the spatial maximum of $2m/r$ and a linear interpolation between consecutive time steps bracketing its first crossing of one. The archived event table is

N_r	h	t_H	r_H	$\Delta M/M_0$
600	0.0500000	4.7133579	3.0518959	$-1.1205\,10^{-2}$
800	0.0374844	4.7162491	3.0506543	$-1.6427\,10^{-2}$
1000	0.0299800	4.7059219	3.0766099	$-1.0696\,10^{-2}$
1200	0.0249791	4.7004729	3.0905271	$-7.6261\,10^{-3}$
1600	0.0187305	4.6952990	3.1040860	$-4.6604\,10^{-3}$
2000	0.0149825	4.6930194	3.1101678	$-3.3322\,10^{-3}$
2400	0.0124844	4.6918209	3.1134115	$-2.6241\,10^{-3}$

The coarsest two event coordinates are not monotone with refinement. On the six finest resolutions the retained nonlinear fits are

$$t_H(h) \simeq 4.6893522370 + 33.88577338\, h^{2.173853282}, \tag{209}$$

$$r_H(h) \simeq 3.1204965744 - 65.59310603\, h^{2.084394965}. \tag{210}$$

These fit exponents are consistent with second-order event convergence on this bounded family. They are not a convergence proof, a reliable error bar for every observable, or a critical-collapse exponent.

For $N_r = 2400$, the interpolated event has

$$t_H = 4.69182094580, \quad r_H = 3.11341154797, \quad m_H = r_H/2 = 1.55670577399. \tag{211}$$

The relation $m_H = r_H/2$ is imposed by the interpolated marginal-sphere criterion. It is not an independent numerical conservation test.

The stored event fields come from the next completed step, at $t = 4.69206969694$, rather than exactly the interpolated t_H. Their recorded outer mass is 1.72538868709, compared with initial outer mass 1.72992821318, a fractional change of $-2.624112\,4\,10^{-3}$. The same stored fields have minimum lapse 0.999829382946, lapse at the sampled peak 1.02582462096, and maximum absolute scalar Ricci curvature 7.35797274672. This timing distinction is small but matters when those quantities are compared with an independently interpolated event.

The scalar curvature diagnostic follows the rescaled convention

$$\mathcal{R} = 2(\Phi^2 - \Pi^2). \tag{212}$$

Finite lapse and curvature values show that this calculation can reach the marginal-sphere event without a coordinate blow-up. They do not provide continuation to a singularity or control of subsequent trapped-region evolution.

19.4 Control and independent diagnostics

The weak pulse on $N_r = 1200$ remains untrapped through $t = 5$, with maximum compactness 0.146722872367 and fractional outer-mass change $-4.5106042\,10^{-3}$. It is a finite-time control in an excised domain. It does not establish global dispersion, passage through a regular center, or a critical amplitude separating two complete physical outcomes.

At the smooth pre-horizon time $t = 3$, the $N_r = 1200$ retained relative residuals are

Diagnostic	Relative residual
Radial mass constraint	$2.27319745\,10^{-6}$
Lapse constraint	$2.80774843\,10^{-6}$
Local mass flux against a time difference	$1.26971871\,10^{-3}$

The flux diagnostic is independent of simply inserting the geometry solve's own right side into its output. Its finite time difference is still an approximation. The separately reported absolute outer-mass flux-balance defect is 0.00230311127. A small total outer-mass change cannot be attributed solely to physical boundary flux unless that flux is integrated and its numerical balance is controlled.

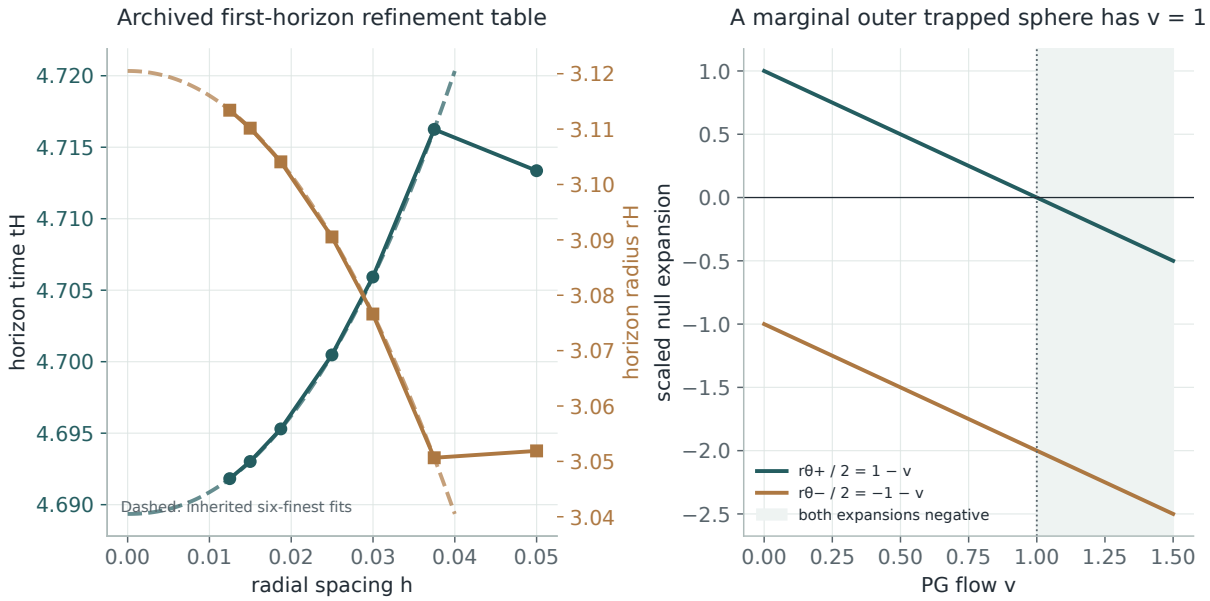


Figure 8. Archived event coordinates under radial refinement, together with the exact expansion and characteristic sign change at $v = 1$. The event table is replotted from retained data. No new scalar-collapse evolution was performed for this figure.

The scalar verification record contains twelve passed checks. Two concern symbolic identities. The remaining checks are bounded numerical tests with declared thresholds. The symbolic audit, null-channel algebra, and seven-row data structure have separate supporting records. Those checks do not constitute a replay of the full collapse or an authentication of its original execution environment. The verification appendix states that evidence boundary.

20 What remains beyond these strong-field sectors

There are exact results here: the electrovac mass identity, its regular-patch constraint equivalence, the reconstructed finite reduced constraints, the scalar flat-slice reduction, the continuum local mass flux, and its compatibility with the radial mass constraint. Their proofs use the displayed chart and symmetry hypotheses. Neither numerical rounding nor a missing fresh Lean build changes them into a machine-checked unrestricted field theory.

There are also controlled numerical observations: a prescribed electrovac gauge trajectory crossing its known outer horizon, and an excised spherical scalar experiment reaching a first marginally trapped sphere. Their retained refinement data and negative control make useful, bounded tests. The next proofs must address the actual boundary closure, regularity margins, vacuum extension, solver conventions, and errors in the full coupled stage map.

For electrovac, a nonvacuous interval energy theorem must control boundary flux and then establish the stronger norm needed by the reconstruction estimate. A full RK4 theorem needs a variable-coefficient argument, rather than the frozen advection symbol alone. For scalar matter, the mass and lapse prefix solves need stability and consistency estimates coupled to the characteristic evolution, including behavior near $v = 0$ and the excision boundary. Event convergence needs quantitative control

of the compactness maximum and its temporal crossing. A tangential spatial root cannot inherit a simple-root horizon bound automatically.

Beyond these radial sectors, finite-range nonlinear gravitational closure remains a separate problem. Spherical symmetry removes the radiative gravitational degrees of freedom. A general $3+1$ Einstein–Maxwell–Dirac construction must restore them, choose boundary and gauge conditions, establish the relevant constraint algebra and chart regularity, and prove consistency and stability of its actual evolution. Numerical marginal-sphere formation here establishes no global cosmic-censorship result, singularity theorem, nonspherical stability theorem, or unique event-horizon reconstruction.

The exact quasienergy chapter in the source corpus supplies a further boundary of interpretation. Its componentwise momentum change of variables conjugates a branchwise nonlinear Lorentz action to ordinary Lorentz transformations. The resulting momentum-space metric is a pullback of a flat metric. This is a kinematic coordinate construction, not a solution of the gravitational constraints. Likewise, positive curvature of a low-momentum dispersion relation is not spacetime curvature. A gravity coupling of the quasienergy system requires its own tetrad, connection, stress-energy, and dynamical constraint analysis. That obligation is not filled by relabeling a momentum reparametrization as strong-field geometry.

21 Constraint structure and the limits of reconstruction

The gravitational branch now has a continuous account from transport geometry to constrained dynamics and spherical tests. The geometric examples identify the additional assumptions needed when a propagation rule is interpreted as gravity. Reduced Gowdy evolution and finite linearized ADM theory make constraint propagation concrete. The nonlinear product-rule obstruction then identifies a specific limitation of a finite pointwise discretization.

The alternatives address that limitation in their own settings. Matrix moment maps retain exact finite algebraic closure. Homological transfer requires a genuine fine differential and controlled contraction data. Resolved quadrature retains the stated polynomial identities under its degree budget, while nonlinear evolution can require additional spectral bands. These constructions therefore carry distinct state spaces and comparison problems.

The Lorentzian classical action supplies a gauge-theoretic parent. Its boundary description and local second-class reduction require the declared regularity, nondegeneracy, canonical transformation, and invertibility hypotheses. Projected discrete-gradient evolution preserves the stated autonomous invariants for an exact step solve within that chart. None of these hypotheses can be discarded when changing coordinates or moving toward a singular configuration.

Spherical mass and flux identities provide unusually direct checks. Electrovac benchmarks test a prescribed slicing trajectory of a fixed charged geometry. The scalar-collapse implementation adds evolved matter and radial constraint solves. Its trapped-sphere diagnostic is a bounded numerical result for the declared data and implementation. Establishing an event in a controlled continuum solution requires an additional existence and convergence argument.

21.1 Mathematical work that remains

A useful next nonlinear theorem should identify a compatible sequence of state spaces and maps, give a level-uniform stability estimate, and state the topology in which the constraint and evolution defects tend to zero. Exact evaluation at one resolution does not itself provide that sequence. A reduction

theorem that crosses charts must also control degeneration of the gauge-fixing and Dirac matrices, boundary conditions, and the change of canonical variables.

The strong field route needs estimates adapted to its actual constraint solve and boundary closure. For scalar matter, this includes the nonlocal radial dependence of reconstructed mass and lapse, the branch of the slicing variables, and convergence strong enough to control the trapped-sphere diagnostic. Extending the construction to nonspherical geometry introduces gravitational degrees of freedom absent from a spherical model.

The inherited massive fermion conventions also need a physical comparison in flat space. Lorentzian signature, the Clifford relation, the kinetic factor, and the adjoint must be matched together. The algebraic gauge model does not settle that normalization.

A quantum theory requires further choices and arguments. The classical master equation does not define a quantum measure or settle anomalies. A common quantum geometry–matter construction and a controlled semiclassical comparison remain research obligations. The verification companion lists these obligations beside the source results they extend.

21.2 From geometric to arithmetic refinement

The next chapter develops register factorization, Haar packets, spectral reduction, and infinite-depth quantum hierarchies. It changes the model while preserving the discipline used here: finite identities, refinement compatibility, and analytic limits require separate arguments. The covariant fiber and cosmology chapters then examine whether a chosen refinement sector admits a physically controlled production history and consistent observable transport.

> **A concrete next comparison**
>
> Name the phase space, constraints, evolution, boundary conditions, and map between resolutions. State the regularity and chart on which the map is defined. Then prove the stability and convergence needed for the physical conclusion being sought.

Sources and further reading

Project sources

[P1] Komorebi P. *Projector Refinement Physics, Volume I*. The source studies cover weak fields, holonomy, Gowdy reduction, linearized ADM constraints, nonlinear transfer, resolved connections, the Lorentzian parent, regular reduction, electrovac geometry, and scalar collapse.

[P2] Komorebi P. *Projector Refinement Physics, Volume I: Technical Supplement*. The nonlinear and strong-field claim dossiers specify boundary, spectral, and numerical hypotheses. The verification appendix maps these hypotheses to the constructions used here.

[P3] Komorebi P. *Komorebi Physics formal source*. Its projector, gauge, and free Dirac results provide related mathematical infrastructure. They do not constitute a compiler certificate for the nonlinear gravity and collapse statements.

Primary context

[C1] R. Arnowitt, S. Deser, and C. W. Misner. "The Dynamics of General Relativity." In L. Witten, ed., *Gravitation: An Introduction to Current Research*, Wiley, 1962, Chapter 7, pp. 227–265. Author republication arXiv:gr-qc/0405109v1, 19 May 2004. https://arxiv.org/abs/gr-qc/0405109v1.

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[C6] R. I. McLachlan, G. R. W. Quispel, and N. Robidoux. "Geometric integration using discrete gradients." *Philosophical Transactions of the Royal Society A* **357** (1999), 1021–1045. https://doi.org/10.1098/rsta.1999.0363.

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[C8] J. Ziprick and G. Kunstatter. "Numerical study of black-hole formation in Painlevé–Gullstrand coordinates." *Physical Review D* **79** (2009), 101503(R). Author preprint *Spherically Symmetric Black Hole Formation in Painlevé–Gullstrand Coordinates*, arXiv:0812.0993v1, 4 December 2008. https://arxiv.org/abs/0812.0993v1; https://doi.org/10.1103/PhysRevD.79.101503.

How these sources are used

The ADM republication supplies the canonical setting for lapse, shift, and constraints. Ashtekar and Batalin–Vilkovisky supply the historical connection and master-equation settings. These references do not establish the projector model's continuum convergence or its quantum anomaly cancellation.

The two Cattaneo–Mnev–Reshetikhin preprints provide the precise comparison for boundary reduction and BV elimination. Section 3.1.3 of arXiv:1201.0290v3 constructs boundary data through a presymplectic quotient and its projectability conditions. Bulk gauge invariance alone therefore does not discharge the missing regular boundary construction in the present gravity model. In arXiv:1507.01221v2, Theorem 2.9 and Corollary 2.12 give BV pushforward and preservation of the quantum master equation for compatible finite-dimensional odd-symplectic data. The identity is conditional on the fine quantum master equation and well-defined integration. It does not manufacture an anomaly-free quantum gravity measure.

McLachlan–Quispel–Robidoux is the established discrete-gradient context. The preservation identity used here is proved directly from the segment integral and skewness of the endpoint Dirac tensor. Misner–Sharp identifies the spherical mass tradition. The charged mass-coordinate identity in this volume is independently differentiated in the stated canonical convention. Ziprick–Kunstatter is prior work on scalar collapse in Painlevé–Gullstrand coordinates. Its regular-center and critical-collapse analyses are distinct from the excised, first-event benchmark retained here.

Relevant full-text sections of the ADM, both Cattaneo–Mnev–Reshetikhin, and Ziprick–Kunstatter author preprints were inspected in their stated versions. For Ashtekar, Batalin–Vilkovisky, McLachlan–Quispel–Robidoux, and Misner–Sharp, bibliographic metadata was authenticated, but publisher full-text access was unavailable. Their uninspected results are not used to discharge a proof obligation. The source record describes this focused comparison and its limits. No priority claim or exhaustive novelty review is made.